NUMERICAL AND EXPERIMENTAL MODAL ANALYSIS ON WASHER DRUM PULLEY

A Project report submitted in partial fulfilment of requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING

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CERTIFICATE

This is to certify that the project entitled **"MODAL ANALYSIS ON WASHER DRUM PULLEY"** is a bonafide work carried out by **Miss. MD.S.GULSHAN BANU, Miss. YAMINI PADIMI, Mr. R.GOKUL SATYA SRIRAM, Mr. ASHRITH BATTU, Mr. B.V.D. PRASAD RATNALA,** in the partial fulfilment of requirements for the award of the degree **Bachelor of Technology in Mechanical Engineering** submitted to the Department of Mechanical Engineering, Andhra University College of Engineering (A), Andhra University, Visakhapatnam during the year 2017-2021 under the guidance of **Dr. G.RAMBABU, Ph.D, Assistant Professor, Dept of Mechanical Engineering**.

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ABSTRACT

The aim of the project work was to perform modal analysis of the pulley, which serves for propulsion of washing machine, and consequently to estimate its modal parameters. There are presented results of the numerical measurements in this project work, verified by experimental method. The end of the project work is devoted to results, comparison and conclusion.

INTRODUCTION

A washing machine is subjected to vibrations due to the dynamic action of centrifugal force produced by the rotating clothes inside it. Due to this, while in operation, at certain load and speed, there is a possibility that anyone of the machine components can reach resonant frequency which can cause failure of that component. At resonance, the amplitude at its natural frequency increases and makes the machine components to vibrate rigorously and fail due to structural damage like large displacements etc.

The motor is coupled to the agitator or the disc and produces rotation. This is achieved by the way of a belt drive between the motor shaft and a pulley connected to the agitator. These are multispeed motors, whose speed can be changed as per the requirement. Hence the pulley is operated in different speeds. In the fully automatic washing machine, the speed of the motor i.e. the agitator changes automatically as per the load on the washing machine.

A washing machine is subjected to vibrations due to the dynamic action of centrifugal force produced by the rotating clothes inside it. Due to this, while in operation, at certain load and speed, there is a possibility that anyone of the machine components can reach resonant frequency which can cause failure of that component. At resonance, the amplitude at its natural frequency increases and makes the machine components to vibrate rigorously and fail due to structural damage like large displacements etc.

A pulley attached to washer drum is a vital mechanical element which rotates the washer drum in the washing machine so that the purpose of that machine is justified. It multiplies the torque generated by the electric motor and produces the amount of torque which is required to rotate the drum. To multiply the torque, it is designed for a considerable radius (greater than motor pulley) which can get deflected and distorted in shape during power transmission. Due to vibrations, if the washer pulley gets damaged, it will result in functional failure of washing machine.

Hence, our sole idea is to perform Modal analysis and Structural analysis on the washer drum pulley and determine the mode shapes and natural frequencies and compare them with the working frequency (rpm) of the washing machine.

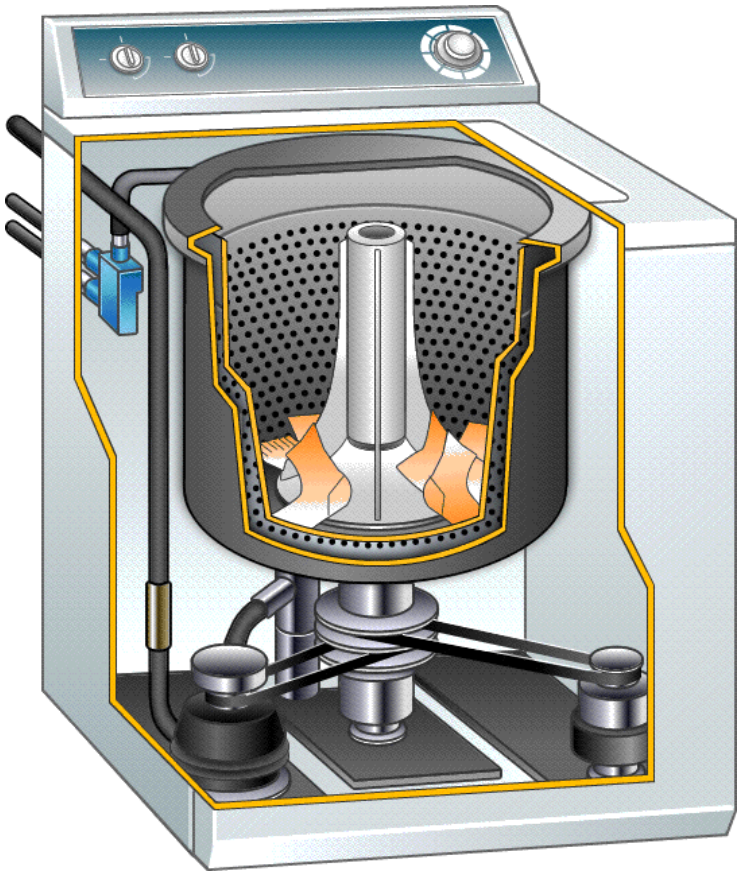


Figure 1.1 Top loaded washing machine (representative image)



Figure 1.2 Various kinds of pulleys

CHAPTER-1

INTRODUCTION TO FINITE ELEMENT ANALYSIS

It is common practice to use approximate solutions of differential equations as the basis for structural analysis. This is usually done using numerical approximation techniques. The most commonly used numerical analyses are:

1. Finite Element Method (FEM):
   Applications: Linear, Non-Linear, Buckling, Thermal, Dynamic & Fatigue analysis
2. Boundary Element Method (BEM): It is most efficient tool to solve acoustics and NVH problems.
3. Finite Volume Methods (FVM): It performs computational fluid dynamics. Unit volume is considered and variable properties at nodes are pressure, velocity, area, mass etc.
4. Finite Difference Method(FDM):

The analysis on washer pulley is linear static analysis and it comes under finite element method.

1.1 Working with FEM:

The origin of the FEM is the need to solve complex elasticity and to make a structural analysis in mechanical engineering. The development can be set back in the 1940s when first approaches to the idea were started. Although the approaches of the pioneers were different, they shared one characteristic: *the mesh of a continuous domain into smaller sub-domains called elements.* These elements were joined together in points called nodes.

FEM is best understood from its practical application, known as finite element analysis (FEA). FEA as applied in engineering is a computational tool for performing engineering analysis. It includes the use of mesh generation techniques for dividing a complex problem into small elements, as well as the use of software program coded with FEM algorithm. In applying FEA, the complex problem is usually a physical system with the underlying physics such as the Euler-Bernoulli beam equation, the heat equation, or the Navier-Stokes equations expressed in either PDE or integral equations, while the divided small elements of the complex problem represent different areas in the physical system.

Almost all applications in the field of mechanical engineering discipline use FEM in the development and design of their products. FEM is a great tool when producing stiffness and strength visualizations and also when optimizing weight, materials and costs. FEM allows the visualization of structures submitted to loads and shows stresses and their distribution. Many different settings can be modified in FEM software in order to obtain the most realistic models. The user, who can manage both

computational time and accuracy required depending on the application, chooses the level of detail. The biggest inconvenience of this method is the blind faith in the results provided by the software. Since it is the user that sets the configuration, it is necessary to be critical with the results and make an intensive discussion before accepting them as valid. FEM has improved the design and the design methodology in many industrial applications, and has reduced the time from idea to production line.

1.1.1 Simulating real-world working environments

To simulate the effects of real-world working environments in FEA, various load types can be applied to the FE model, including nodal (forces, moments, displacements, velocities, accelerations, temperature and heat flux), elemental (distributed loading, pressure, temperature and heat flux), as well as acceleration body loads (gravity).
Types of FE analysis include linear statistics, nonlinear statics and dynamics, normal modes, dynamic response, buckling and heat transfer. Typical results calculated by the solver include nodal displacements, velocities and accelerations, as well as elemental forces, strains and stresses.

1.1.2 Benefits of FEA

FEA can be used in new product design, or to refine an existing product, to ensure that the design will be able to perform to specifications prior to manufacturing. With FEA one can:

- Predict and improve product performance and reliability.
- Reduce physical prototyping and testing.
- Evaluate different designs and materials.
- Optimize designs and reduce material usage.

1.2 NX 11 SOFTWARE:

NX, formerly known as "unigraphics", is an advanced high-end CAD/CAM/CAE, which has been owned since 2007 by Siemens PLM Software. It contains various FEM solvers like NASTRAN, Simcenter Thermal and Flow etc.

1.3 INTRODUCTION TO NASTRAN:

The 1964 annual review of NASA's structural dynamics research program revealed that the research centers were separately developing structural analysis software that was specific to their own needs. The review recommended that a single generic

software program should be used instead. In response, an ad hoc committee was formed. The committee determined that no existing software could meet their requirements. They suggested establishing a cooperative project to develop this software and created a specification that outlined the capabilities for the software. The eventual formal name approved by NASA for the program, NASTRAN, is an acronym formed from NASA Structure Analysis. The NASTRAN system was released to NASA in 1968.

1.3.1 Software architecture

NASTRAN is written primarily in FORTRAN and contains over one million lines of code. NASTRAN is compatible with a large variety of computers and operating systems ranging from small workstations to the largest supercomputers.

NASTRAN was designed from the beginning to consist of several modules. A module is a collection of FORTRAN subroutines designed to perform a specific task—processing model geometry, assembling matrices, applying constraints, solving matrix problems, calculating output quantities, conversing with the database, printing the solution, and so on. The modules are controlled by an internal language called the Direct Matrix Abstraction Program (DMAP).

Each type of analysis available is called a solution sequence.

Some of the most common solution sequence codes are:

- 101 - Linear Static
- 103 – Modal
- 105 – Buckling
- 106 - Non-Linear Static
- 107 - Direct Complex Eigen value
- 108 - Direct Frequency Response
- 109 - Direct Transient Response
- 110 - Modal Complex Eigen value
- 111 - Modal Frequency Response
- 112 - Modal Transient Response

1.4 STEPS IN FINITE ELEMENT METHODS:

The basic steps involved in finite element methods are:-

1) Discretization of the domain.

2) Selection of an interpolation model to represent the variation of the field variable.

3) Derivation of element characteristic matrices and vectors.

4) Assemblage of element characteristic matrices and vectors to obtain the system equations.
5) Solution of system equations to find the field values of the field variables.
6) Computation of element resultants.

For this project, following sequence codes were used:

Sequence code101- Linear Static is used for this project to perform structural analysis.

Sequence code 103-Modal is used for this project to perform modal analysis.

Assumptions:

- A worn-out pulley has a change in its geometry, which could be reduction in mass or stiffness or a change in centre of mass which in turn affects the natural frequency of it.

- During the simulation, the pulley is assumed to be new; hence the Natural frequencies obtained will be very close to that of a new component. For a good designed pulley modal frequency of the pulley must not coincide with the operating frequency of the washing machine.

.

Chapter 2

MESHING

2.1 INTRODUCTION TO MESHING:

To be able to make simulations, a mesh, consisting of up to millions of small elements that together form the shape of the structure needs to be created. Calculations are made for every single element. Combining the individual results gives us the final result of the structure. The approximations we just mentioned are usually polynomial and in fact, interpolations over the element(s). This means we know values at certain points within the element but not at every point. These 'certain points' are called nodal points and are often located at the boundary of the element. The accuracy with which the variable changes is expressed by the approximation, which can be linear, quadratic, cubic etc.

2.2 TYPES OF3-D MESHING ELEMENTS:

There are four different 3D element types — Tetrahedral, bricks, prisms, and pyramids. These four elements can be used, in various combinations, to mesh any 3Dmodel. (For 2D models, you have triangular and quadrilateral elements available)

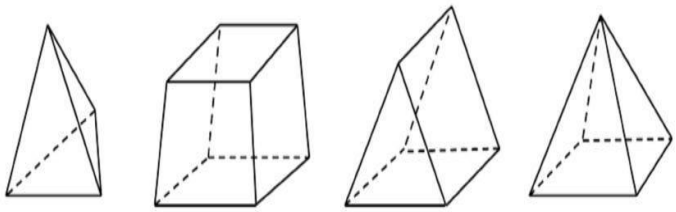


Figure 1.3 Tetrahedral, Brick, Prism, Pyramid

Tetrahedral are also known as a simplex, which simply means that any 3D volume, regardless of shape or topology, can be meshed with tetrahedral. They are also the only kind of elements that can be used with adaptive mesh refinement. For these reasons, tetrahedral can usually be your first choice. The other three element types (bricks, prisms, and pyramids) should be used only when it is motivated to do so. It is first worth noting that these elements will not always be able to mesh a particular geometry. The meshing algorithm usually requires some more user input to create such a mesh.

CHAPTER 3

STRUCTURAL ANALYSIS ON WASHER PULLEY USING NX NASTRAN

3.1 INTRODUCTION:

Structural analysis is the determination of the effects of loads on physical structures and their components. Every machine element that is designed to carry loads and transfer the motion will undergo some amount of deformation that results in the formation of stresses. This will affect the utility of the component.

Structural analysis on a machine element like washer pulley of a washing machine can be done by simple analytical methods using maximum load conditions if it has a simple geometric shape. But if the geometry varies it would be difficult to carry the tedious hand calculations which is time consuming and limited by the boundary conditions.

A pulley is subjected to ‘spatially varying load’ conditions which means the software tapers the load from point to point according to the difference between the calculated load value at each interpolation point. Because all the load over the pulley rim is not same it is more in the middle of the rim-belt contact region and goes on reducing to the other ends. Spatially varying load is discussed later.

3.2 PROCEDURE:

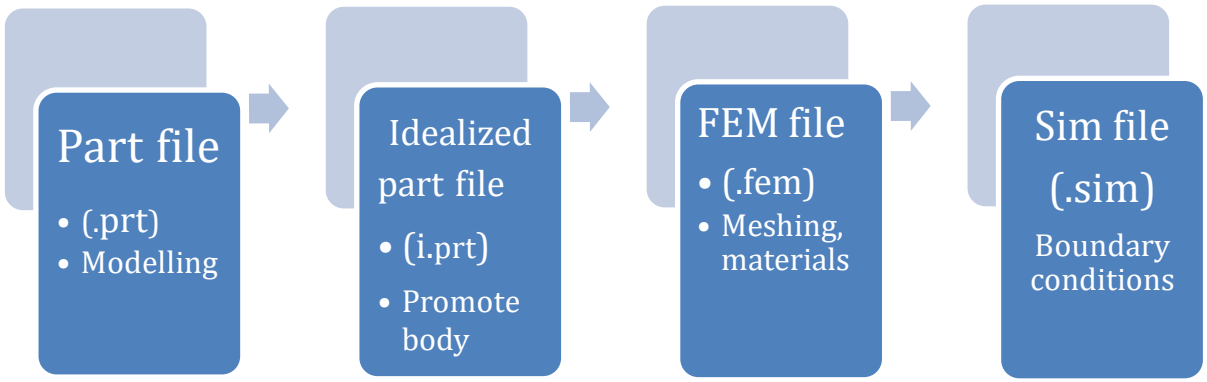


Figure 3.1

3.2.1 Modelling:

The CAD model of the washer pulley was modelled in NX_11 Unigraphicsw.r.t the available component. Various blends are estimated and included in the model. After modelling, the component is splitted into different bodies using ‘split body’ option in home tab > feature group > more > split body.

Splitting the component will be an advantage in applying loads and constraints in ‘sim file’. But it creates the necessity of mesh mating conditions in ‘fem file’.

The pulley rim and spokes are splitted with the help of an extruded surface by making the extruded surface as the reference plane. The extruded surface with translucency is shown in the figure below.

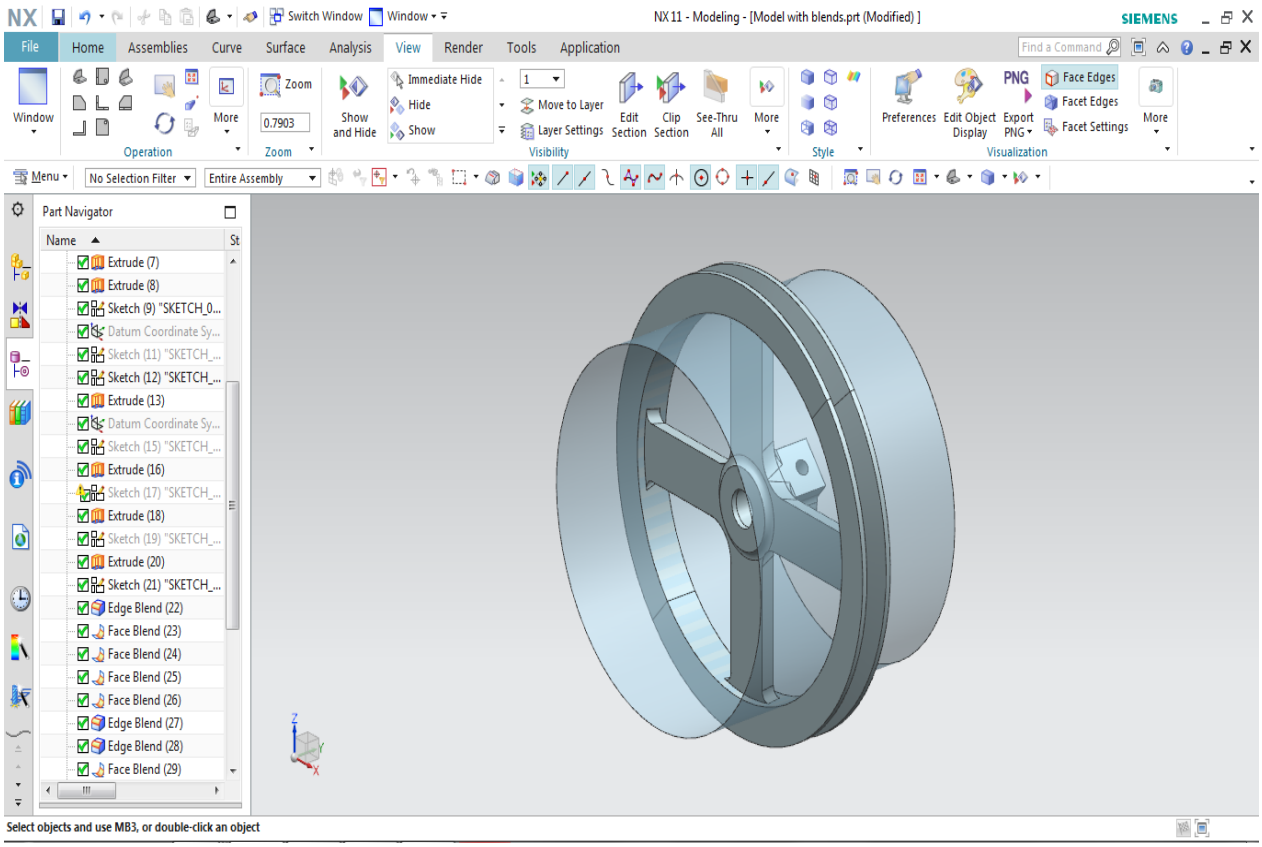

NX
Switch Window
Window
NX 11 - Modeling - [Model with blends.prt (Modified)]
SIEMENS
File
Home
Assemblies
Curve
Surface
Analysis
View
Render
Tools
Application
Find a Command
Window
Operation
More
Zoom
0.7903
Show and Hide
Immediate Hide
Hide
Show
Move to Layer
Layer Settings
Edit Section
Clip Section
See-Thru All
More
Visibility
Style
Preferences
Edit Object Display
Export PNG
Face Edges
Facet Edges
Facet Settings
More
Visualization
Menu
No Selection Filter
Entire Assembly
Part Navigator
Name
Extrude (7)
Extrude (8)
Sketch (9) "SKETCH_0...
Datum Coordinate Sy...
Sketch (11) "SKETCH_...
Sketch (12) "SKETCH_...
Extrude (13)
Datum Coordinate Sy...
Sketch (15) "SKETCH_...
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Sketch (17) "SKETCH_...
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Extrude (20)
Sketch (21) "SKETCH_...
Edge Blend (22)
Face Blend (23)
Face Blend (24)
Face Blend (25)
Face Blend (26)
Edge Blend (27)
Edge Blend (28)
Face Blend (29)
Select objects and use MB3, or double-click an object

Figure 3.2

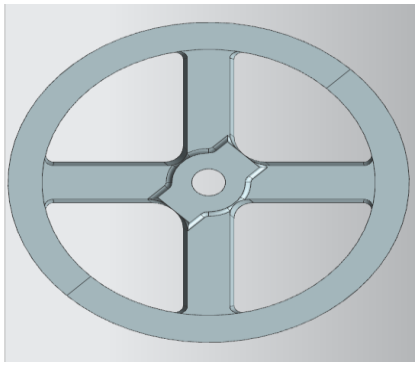


Figure 3.3



Figure 3.4

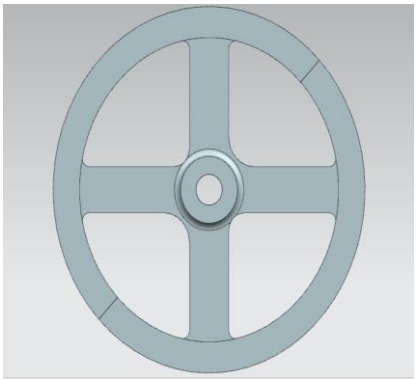


Figure 3.4

3.2.2 Pre/post:

After modelling, go to Application tab > simulation > pre/post to start analysis on the modelled component.

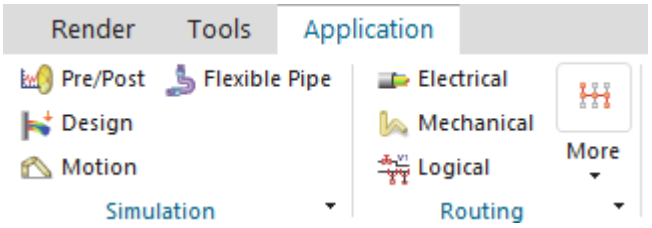


Figure 3.5

3.2.3 Idealizing:

After pre/post select "New FEM and Simulation" in the context and check the "create idealized part" in the dialog box that appeared

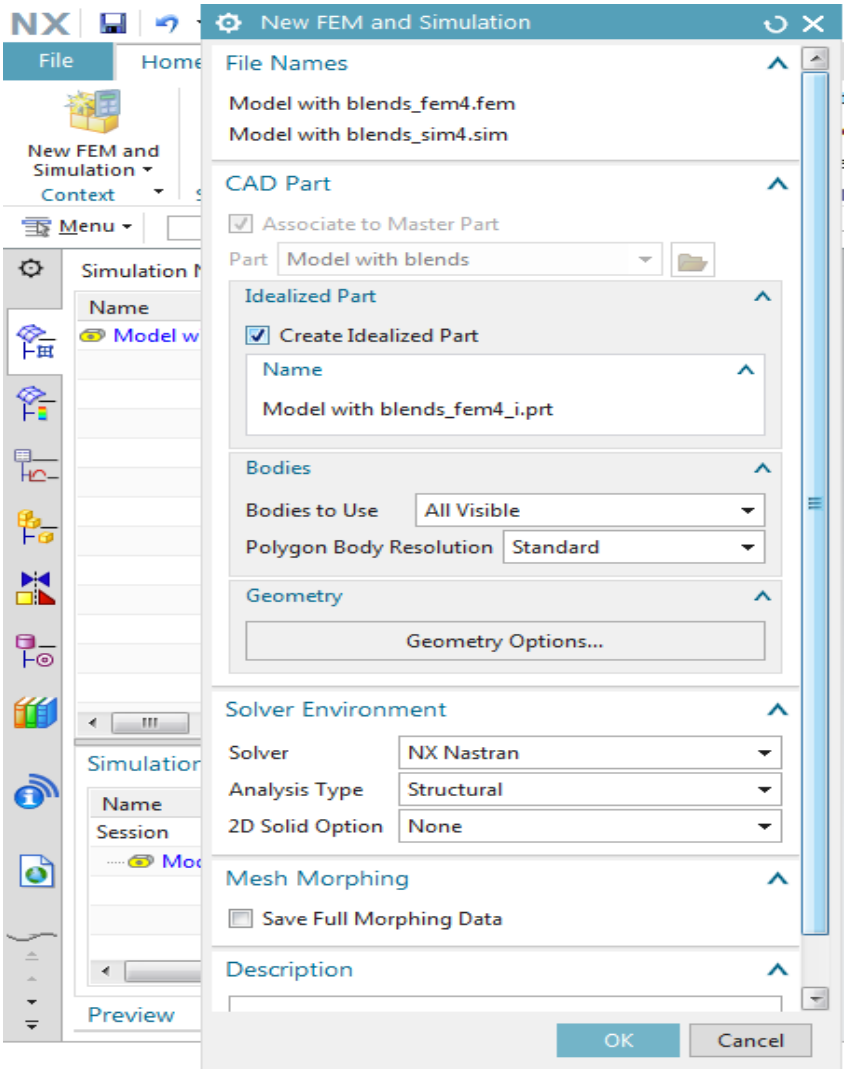


Figure 3.6

The checking will create the idealized part file.

Solver used for the analysis is NX NASTRAN for structural analysis. After clicking OK, solution file is opened.

Solution used for structural analysis is SOL 101 Linear Statics-Global constraints.

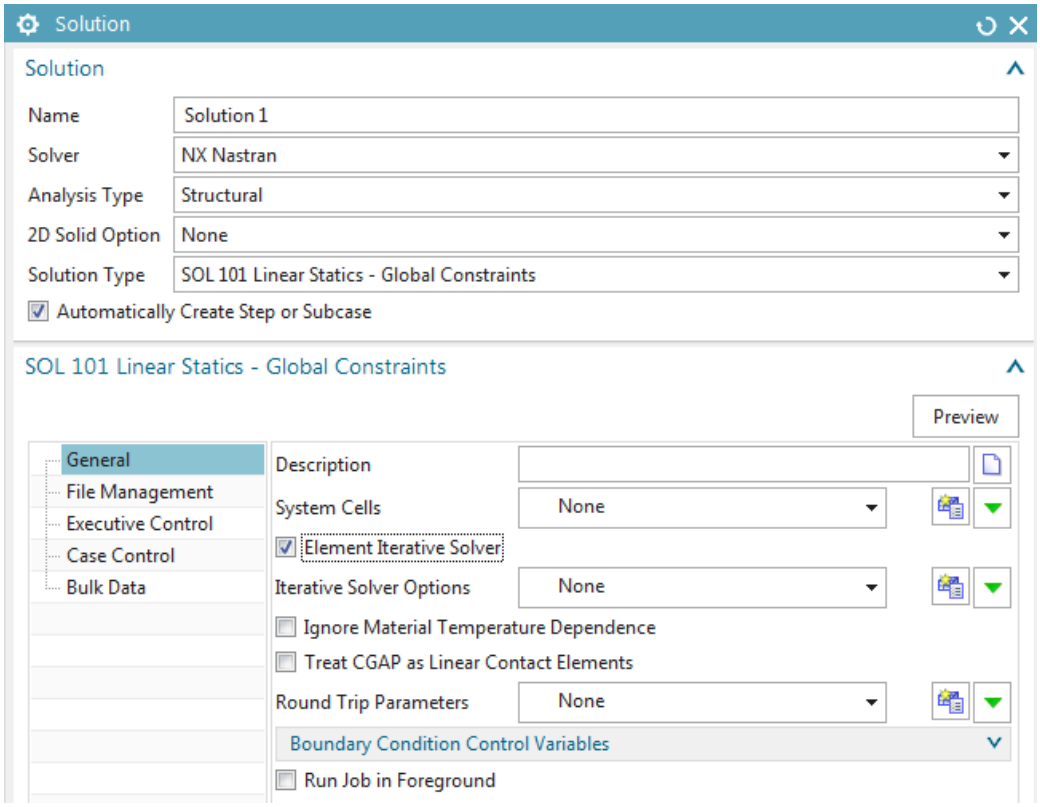


Figure 3.7

New FEM file is opened. Then go to Change displayed part >i.prt file > OK. Now i.prt file is opened.

3.2.4 Promote the body.

Significance of promote body:

To associatively copy a sheet or solid body from the idealized part into a higher level in the assembly hierarchy, use the Promote command.

After promoting the body, we can use the geometry idealization commands and Modeling application commands to add or modify features.

A promoted body remains associated to the history in the master part. If updates occur to the master part, the software also updates the promoted body in the idealized part.

Advantages of using Promote versus WAVE Geometry Linker:

Although WAVE linking geometry can be more robust and versatile, Promote can be simpler and more straightforward to use, particularly if you do not have a lot of experience working with assemblies in this software.

3.2.5 FEM File:

Go back to FEM file through ‘change displayed part’. Or double click on the ‘fem file’ in Simulation file view tab in simulation navigator.

Using seed mesh technique, the component is meshed. Seed mesh technique involves meshing the component first with 2D mesh and then with 3D Tetra mesh on the 2D mesh. The tetra mesh will follow the mesh pattern of the 2D mesh resulting in more uniform mesh. Mesh size has to be controlled properly to avoid Jacobian error.

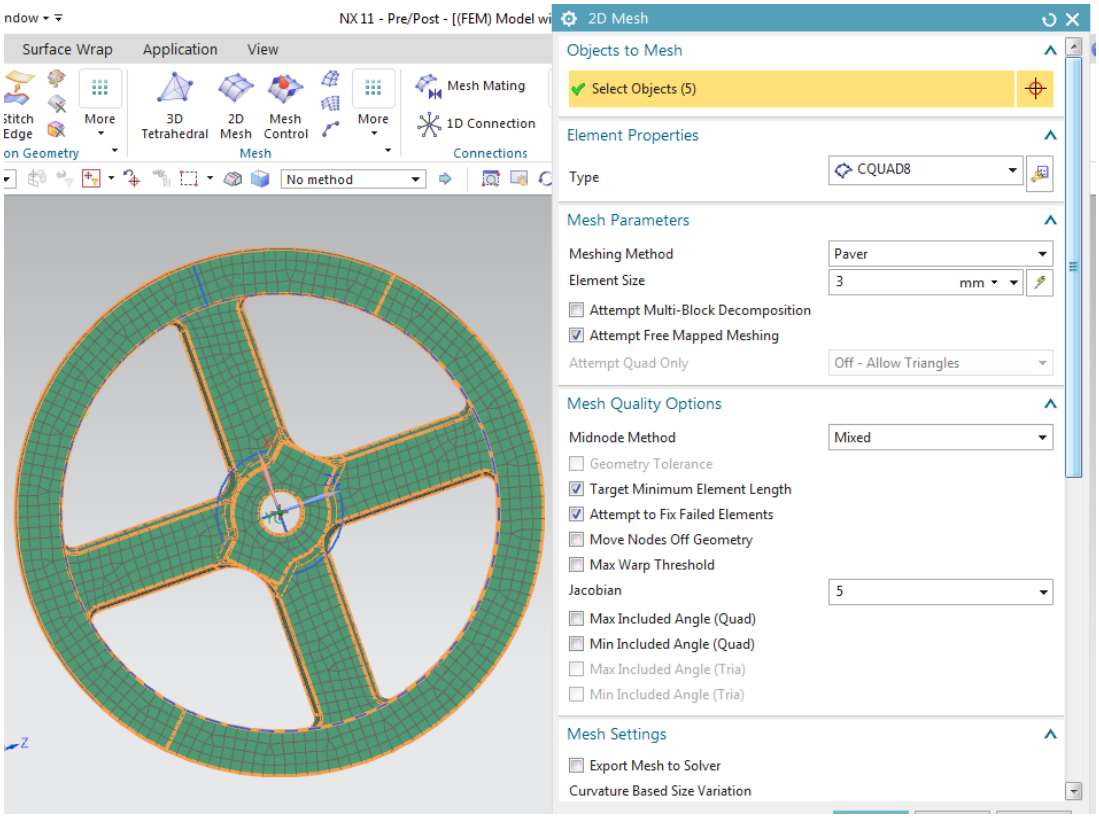


Figure 3.8

2D Meshing:

Element size	3mm
Element Type	CQUAD8
Meshing method	Paver

Uncheck the ‘Export mesh to solver’ as *2D meshing is only for reference for 3D tetra* in seed mesh technique. So, no need for material specification and mesh thickness in 2D meshing.

3D Meshing:

Element size	3mm
Element Type	CTETRA(10)
Material used	Aluminum_A356

Specify the material by disabling ‘Automatic creation’ in destination collector.

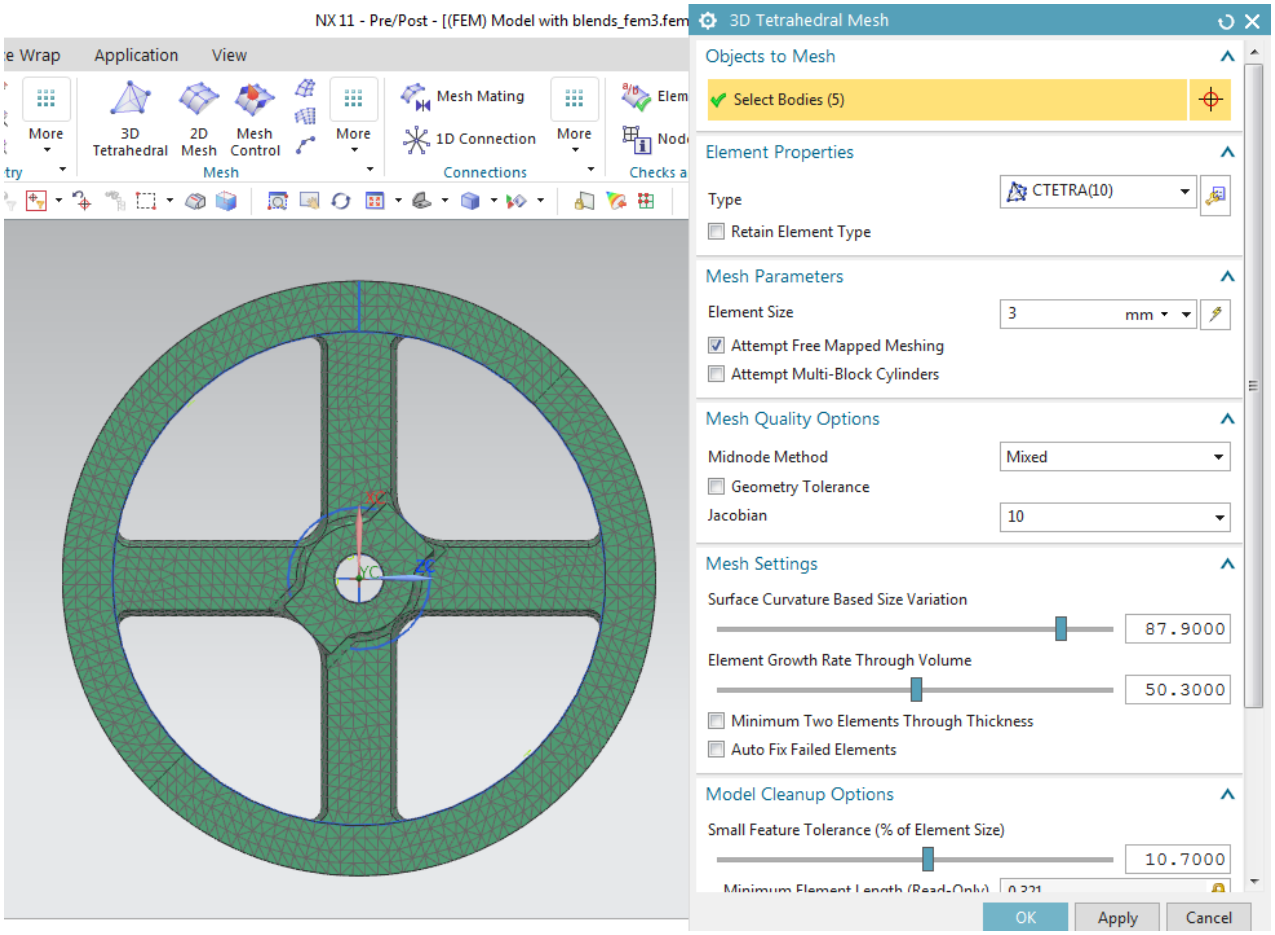


Figure 3.8

Mesh mating conditions:

As the body is splitted, even though the meshing is made, the load will not transfer to the remaining parts. In that case apply mesh mating conditions in the ‘connections’ group. Choose ‘automatic creation’ and ‘Glue Non-coincident’ so that mesh mating conditions will be created on the component. ‘Glue Non-coincident’ is preferred because it won’t disturb the mesh pattern at mating areas.

Disadvantage of going with automatic mesh mating condition is it will create a new collector for every connection and the choice of collector (RBE2/RBE3) is not in our hands unlike in ‘simulation object types’ in ‘sim file’. For our analysis the software created RBE3 collector.

REB2 transfer equal force in all directions but REB3 transfer force according to distance i.e., the node which near to RBE experience more force and vice versa.

3.2.6 SIM File:

Double click on _sim file in simulation file view in simulation navigator to change from ‘fem file’ to ‘sim file’ or use ‘change displayed part’ option in the context.

First of all, the component is constrained at the hub of the pulley using fixed constraint as the pulley is firmly fixed to the washer shaft.

Go to constraints > fixed constraint > select the faces > OK. Edit the display from large towards tiny.

Now, go to load type in loads and conditions select pressure. Belt contacts only with the side faces of the pulley groove. Hence select the two faces for the half portion of the rim.

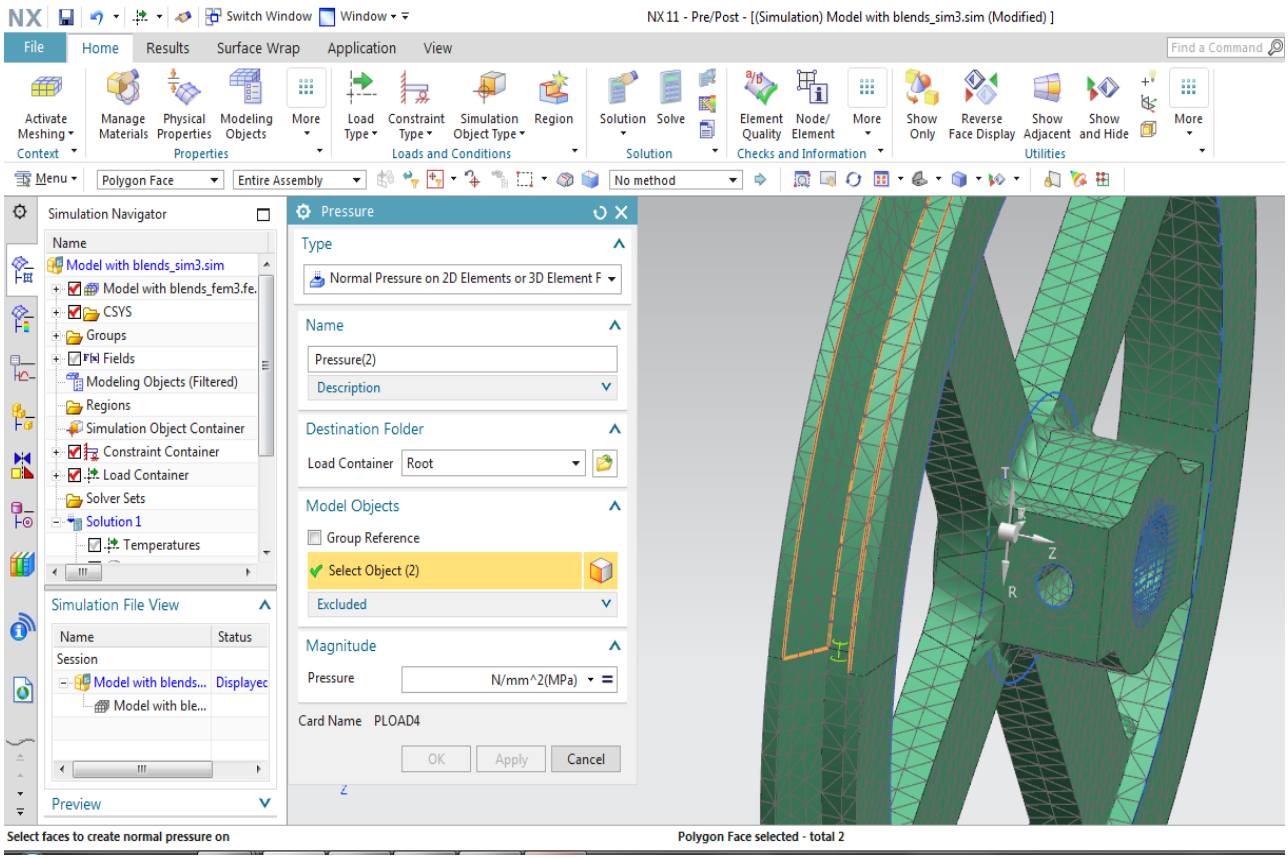


Figure 3.9

Select a new field and choose cylindrical coordinate system as the hub of the pulley is cylindrical. You can create a new cylindrical coordinate system there itself or you can use the existing one if previously created.

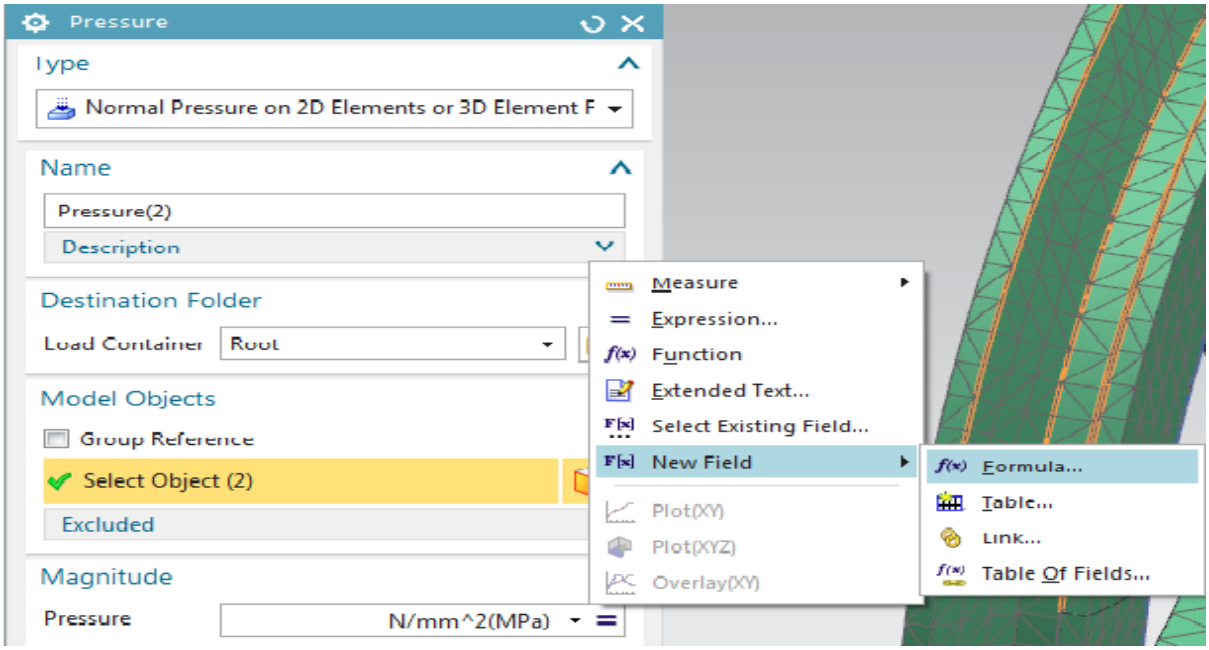


Figure 3.10

Change the domain and spatial map to ‘cylindrical’. Select the cylindrical coordinate system along the axis of the pulley.

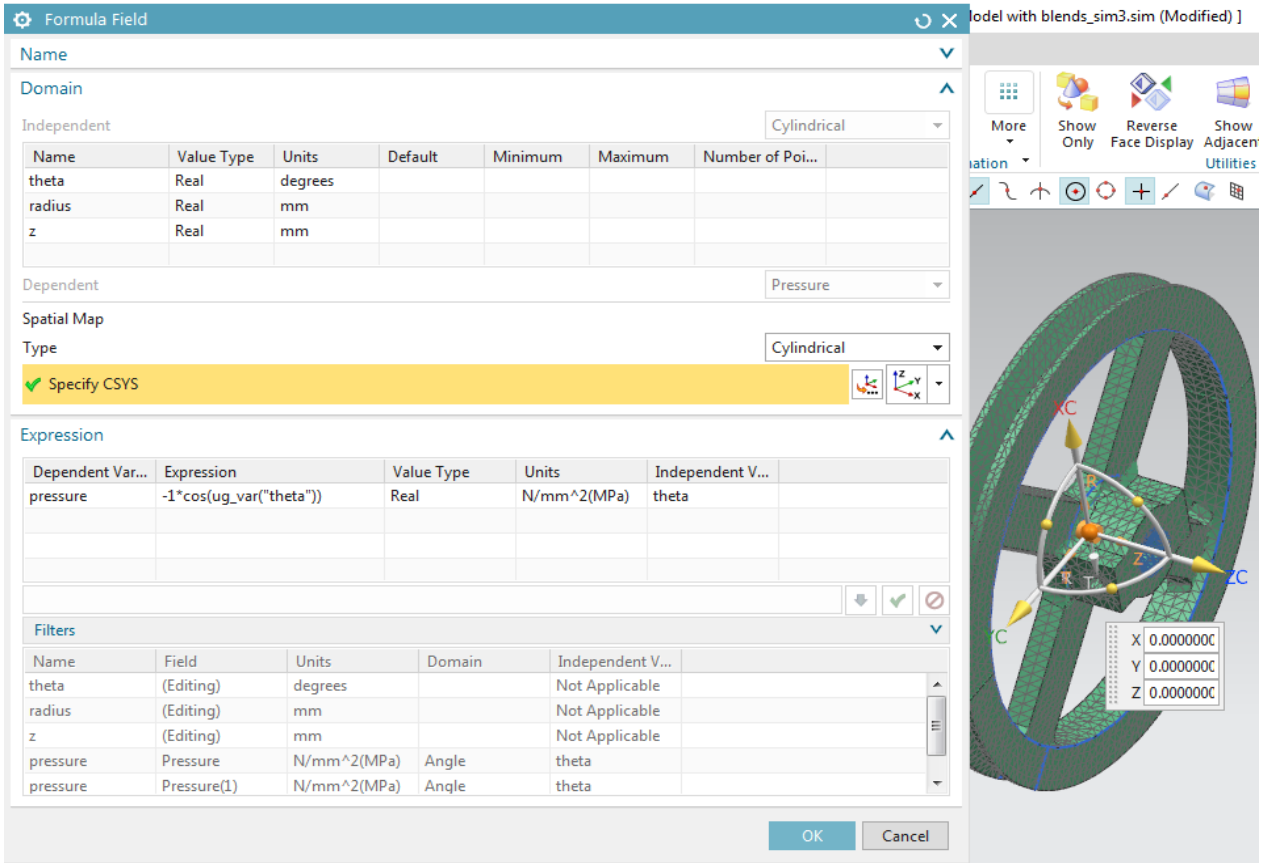


Figure 3.11

The expression given to the above formula field is -1*cos(ugvar("theta")).

Dependent variable	Pressure (N/mm^2)
Independent variable	Theta

- Negative sign is due to the coordinate system direction.

Check the tick box and click OK >OK.

Now go to simulation navigator > Loads > Pressure > Right click > edit display

- Display mode: Expanded
- Scale: Towards ‘Large’

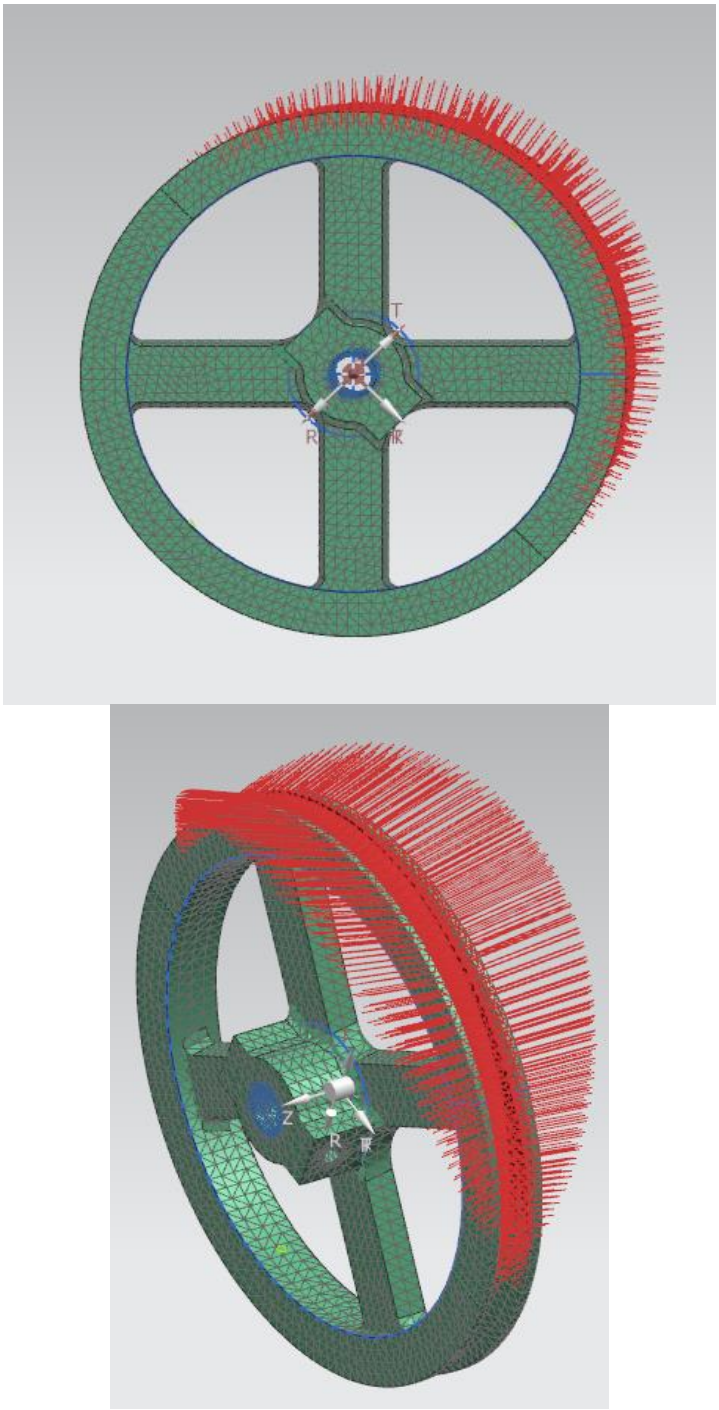


Figure 3.12

The pressure is displayed as uniformly varying over the half of the rim.

NOW SOLVE

3.3 POST PROCESSING:

Post processing gives the simulation results

Case 1: when load is applied on the rim where max. pressure is in between the spokes.

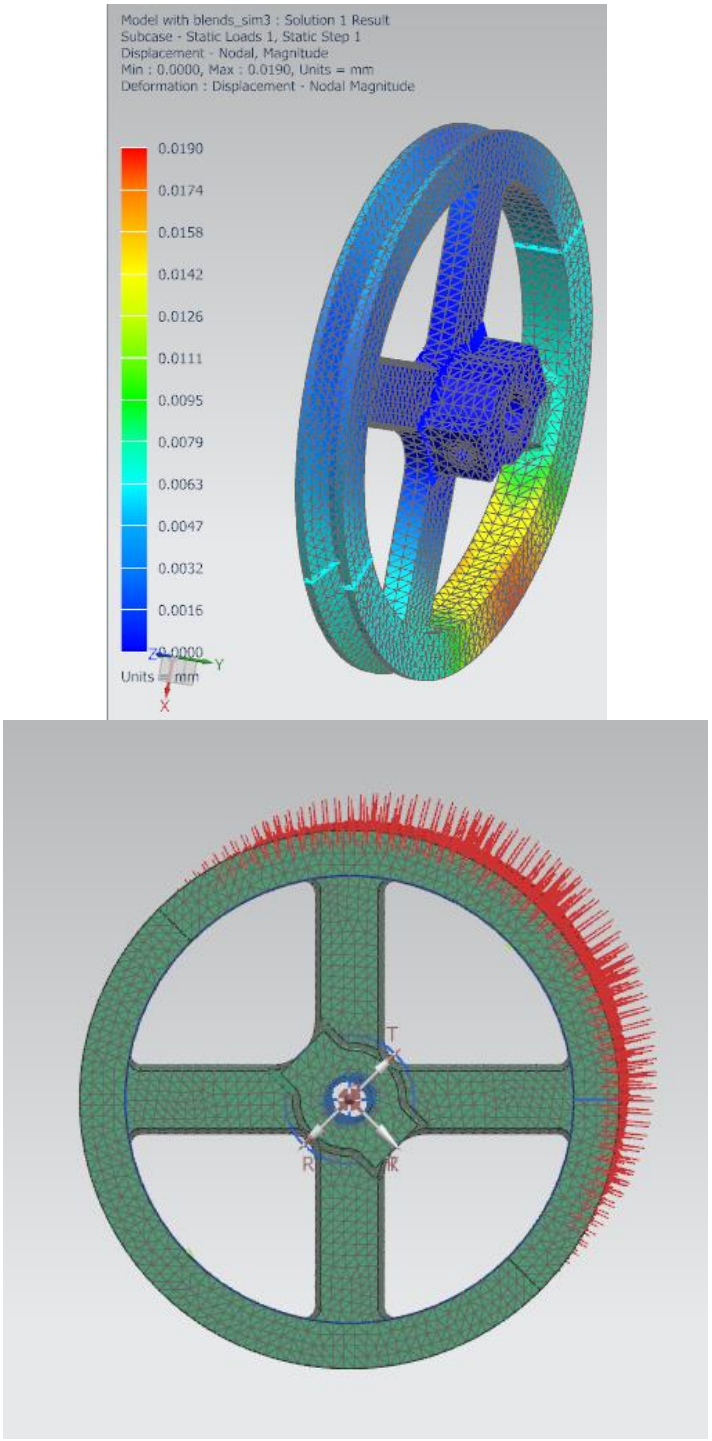


Figure 3.13

Maximum displacement: 0.0190 mm

Stress elemental : 21.4 N/mm^2 (max)

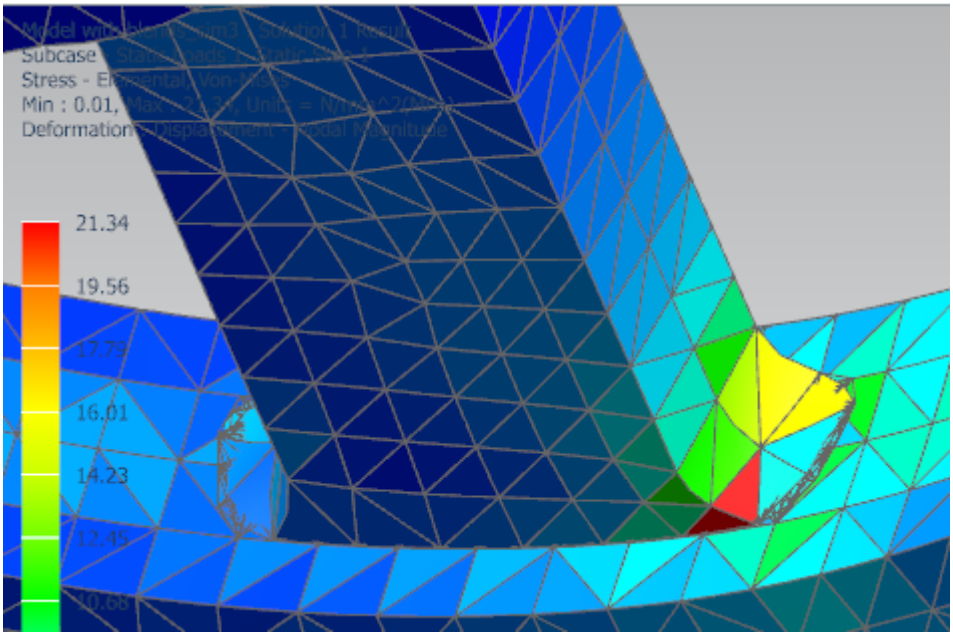
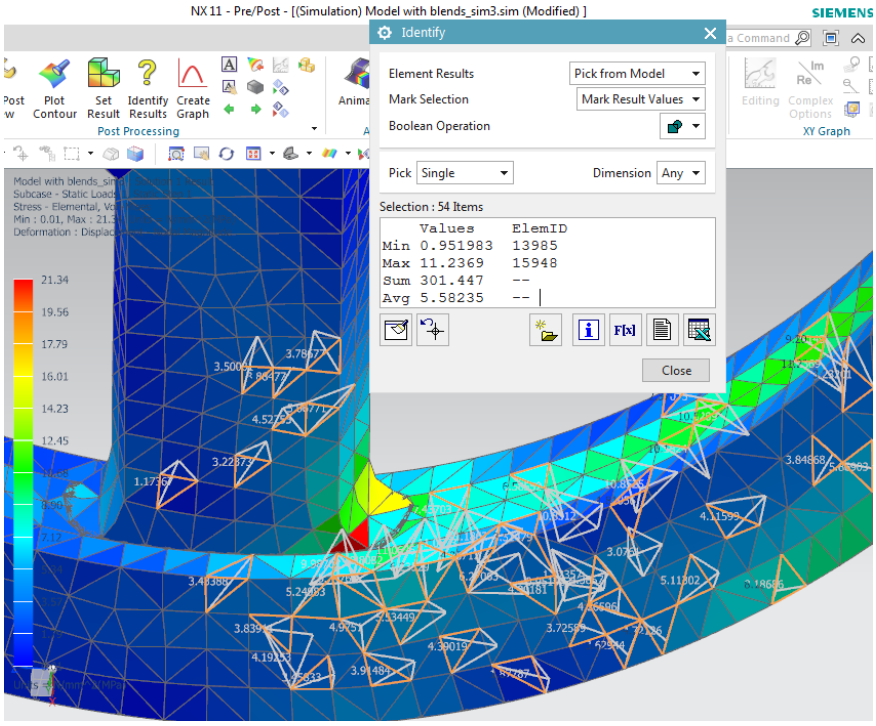


Figure 3.14

2 stress elemental

Here the max stress is resulted at the fillet at spokes and rim contact. But the avg. stress on the pulley on the belt region is 5.58 N/mm^2. (Average stress is the average of stress on the selected elements)

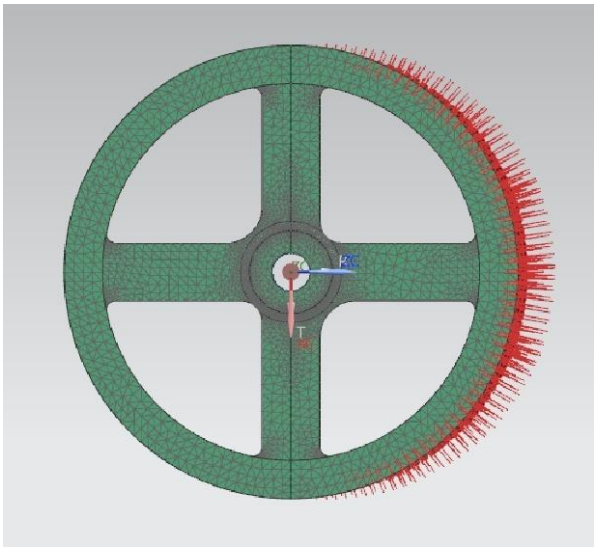


3 avg. stress

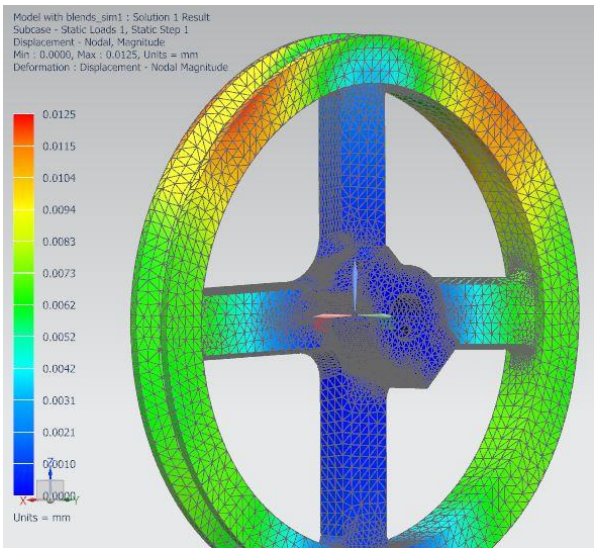
Figure 3.15

From this we can see that as the elements at the fillets are very less and the load is taken by the fillets, they are experiencing more stress.

Case 2: When load is applied on the rim where max. pressure is along the spokes.

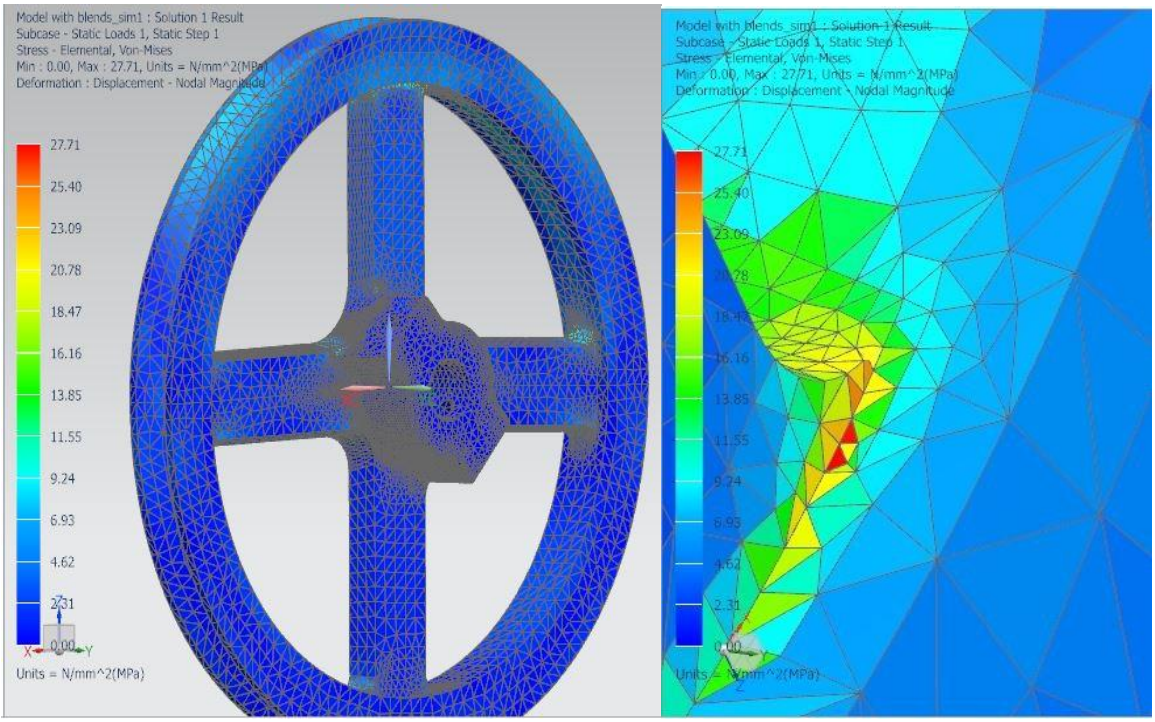


4: case2
Figure 3.16



5: max displacement in case2
Figure 3.17

Max. Displacement: 0.0125mm



6: Stress elemental

Figure 3.18

Stress Elemental : 27.71 N/mm^2 (max)

Here the max stress is resulted at the fillet at spokes and rim contact. But the avg. stress on the pulley on the belt region is 5.87 N/mm^2. (Average stress is the average of stress on the selected elements)

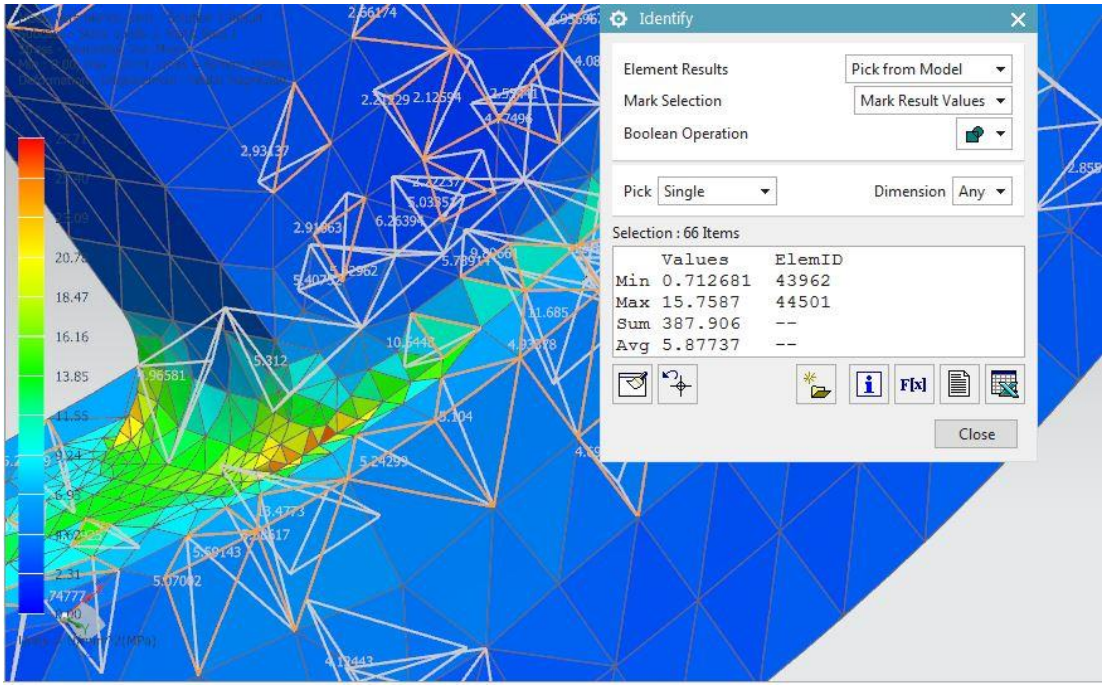


Figure 3.19

In case of overload the failure occurs at the interface of rim and spoke where fillets are not applied.

3.4 RESULT:

The following results were obtained after post-processing.

Case 1: when load is applied on the rim where max. pressure is in between the spokes.

1.	Max. Displacement	0.0190mm
2.	Stress Elemental	27.71 N/mm^2
3.	Average Stress	5.87 N/mm^2

From this we can see that as the elements at the fillets are very less and the load is taken by the fillets, they are experiencing more stress. The max stress is resulted at the fillet at spokes and rim contact.

Case 2 : when load is applied on the rim where max. pressure is along the spokes.

1.	Max. Displacement	0.0125mm
2.	Stress Elemental	21.4 N/mm^2
3.	Average Stress	5.58 N/mm^2

Here the max stress is resulted at the fillet at spokes and rim contact.

3.5 SIMULATION DETAILS

Model Construction

The version of Product used to create this model was NX 11.0.2.7.

Part File names used to construct this model:

File names in session

Part File Name	File Type
Model with blends_sim1.sim	SIM
Model with blends_fem1.fem	FEM

Model Summary:

Solution Summary table

Solution Detail	Status
Solution Name	Solution 1
Solver	NX Nastran
Analysis Linearity	Linear
Solution Type	SOL 101Linear Static Analysis

Mesh Summary table

Mesh Item	Count
Total element count	54499
Total node count	85523
Tet10 element count	54499

Meshes

Node Count	*85523*
Element Count	*54499*

Mesh Summary

Name	Type	Element Count	Node Count	Physical Property Name
3d_mesh(1)	CTETRA(10)	54499	85523	PSOLID1

Materials

Materials Used Summary

Name	Type	Referenced By
Aluminum_A356	Isotropic	PSOLID1 (Phys Prop)

CHAPTER-4

INTRODUCTION TO VIBRATIOINS

4.1 FREE VIBRATIONS:

The basic vibration model consists of a mass, spring (stiffness) and damper (damping).

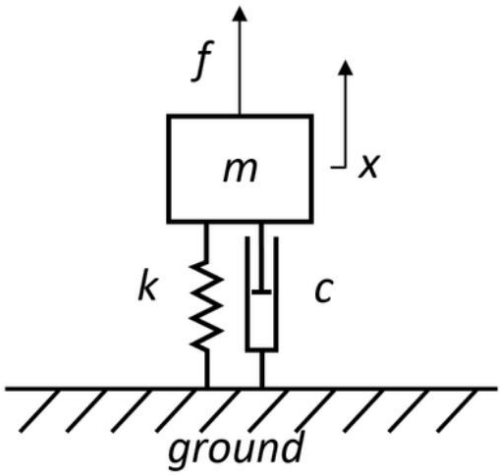


Figure 4.1

In the absence of damping, the system is considered conservative and principle of conservation of energy offers another approach to the calculation of the natural frequency.

When the system is disturbed from its initial position, the resulting free-vibration of the system will be a combination of the different normal modes. The participation of different modes will depend on the initial conditions of displacements and velocities. So, for a system the free vibration can be given by

$$x = \phi_1 A \sin(\omega_1 t + \psi_1) + \phi_2 B \sin(\omega_2 t + \psi_2)$$

Where, ψ_1 and ψ_2 are phase difference

4.2 NATURAL FREQUENCY AND RESONANCE:

Natural vibrations are different from forced vibrations which happen at frequency of applied force (forced frequency). All physical structures have natural frequencies due to natural vibrations. These are the frequencies at which the structure will tend to vibrate when subjected to certain external forces. These frequencies are dependent on the way mass and stiffness are distributed within the structure.

The natural frequency is an inherent property of the object. There are only two ways in which the natural frequency can be changed: either change the mass, or change the stiffness.

$$\omega_n = \sqrt{k/m}$$

Figure 4.2

Resonance:

Resonance is a phenomenon in which a dynamic force drives a structure to vibrate at its natural frequency. When a structure is in resonance, a small force can produce a large vibration response.

When a dynamic force is applied to a physical object, it will vibrate. When a force is applied at the object's natural frequency, it goes into resonance, and a higher amplitude vibration response is created.

4.3 MODAL ANALYSIS:

The determination of the natural frequencies and modes of vibration of a system is known as modal analysis. A mode of vibration is characterized by vibration at a particular frequency (natural frequency) of all points of the system.

A modal analysis determines the vibration characteristics (natural frequencies and mode shapes) of a structure or a machine component. It can also serve as a starting point for another, more detailed, dynamic analysis, such as a transient dynamic analysis, a harmonic analysis, or a spectrum analysis. The natural frequencies and

mode shapes are important parameters in the design of a structure for dynamic loading conditions.

Modal testing also combines data acquisition with further analysis. In an industrial application, the complete process is often referred to as modal testing and analysis, or experimental modal analysis (EMA).

Results of modal testing and analysis are used in various simulation and testing applications, including vibration response calculations, root cause analysis of vibration problems and damage detection, but also for adding flexibility to multibody analysis and speeding up durability and vibro-acoustic simulations. Modal-based calculations are very effective and allow efficient evaluation of structural changes to responses of any kind.

4.4 REAL EIGEN VALUE ANALYSIS/ UN-DAMPED NATURAL FREQUENCY ANALYSIS:

The first usual step in performing a dynamic analysis is determining the natural frequencies and the mode shapes of the structure with damping neglected. The results characterize the basic dynamic behaviour of the structure and are an indication of how the structure will respond with dynamic loading.

The deformed shape of the structure at a specific natural frequency of vibration is termed its normal mode of vibration. Some other terms used to describe the normal mode are:

- Characteristic shape
- Fundamental shape
- Mode shape

Each mode shape is associated with a specific natural frequency.

Natural frequencies and mode shapes are functions of the structural properties and boundary conditions. A cantilever beam has a set of natural frequencies and associated mode shapes. If structural properties change, the natural frequencies change but the mode shapes may not necessarily change.

Reasons to compute normal modes

- To assess the dynamic interaction between a component and its supporting structure.
- To determine the operating frequency of the rotating components and the components in an assembly or structural components attached to it.
- If the frequencies are close which during the operation may lead to structural damage or failure.

4.5 MODE SHAPES:

The SDOF example system had one natural frequency. Structures in the real world are more complex, and have multiple degrees of freedom (MDOF). As a result, real world structures have many natural frequencies. The structure vibrates differently at each of these natural frequencies. How it moves at a particular frequency is called a mode shape.

Mode shapes give valuable insight into how a structure behaves when operating at its natural frequencies. The shape can show the engineer where to constrain/modify a structure to reduce the vibration response, or how to shift the natural frequency so it does not coincide with the frequency of an excitation.

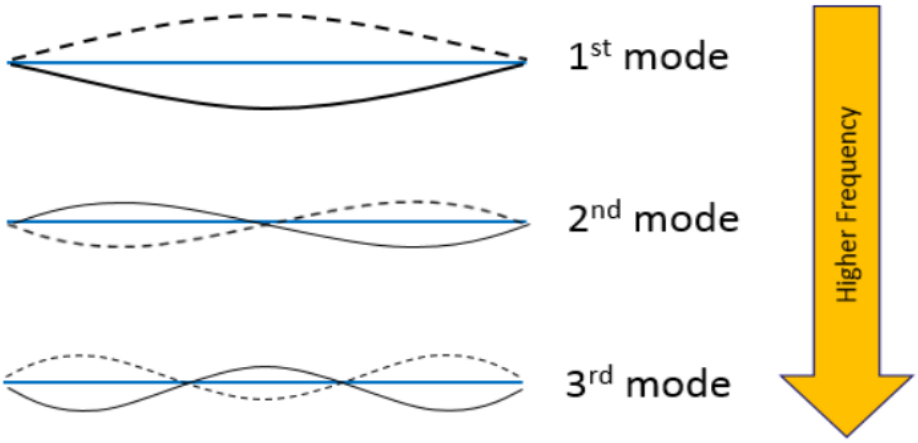


Figure 4.3

At higher frequencies usually modes become local in natural, rather than global. In a global mode, the entire structure participates while in a local mode, only part of the structure participates.

4.6 BENEFITS OF MODAL ANALYSIS:

Modal analysis through FE analysis or structural testing.

Modal analysis performed through either structural testing or finite element analysis-based simulation helps you:

- Understand how a structure vibrates
- Correlate and update simulation models
- Speed up structural, vibro-acoustic and durability calculations
- Include flexibility in multi body simulation models.

CHAPTER-5

MODAL ANALYSIS ON WASHER DRUM PULLEY

Here during simulation, the body is considered to be having no damping co-efficient hence the natural frequencies of the component are extracted with this condition.

5.1 LANCZOS:

Lanczos Performs real eigenvalue analysis on real symmetric mass and stiffness matrices using the **LANCZOS** method for the eigen solution and Lagrange Multiplier techniques for constraint processing.

The Lanczos method normalizes the computed eigenvectos using the mass or max method.

Frequency Range	Lower Limit (Hz)	Upper Limit (Hz)
	0	4096

The Frequency Range is selected according to the operating condition of pulley in the washing machine. Our frequency range of interest is below 4096Hz. Selecting this range limits the modes the solver calculates according to the frequency of each mode.

Number of desired modes	100

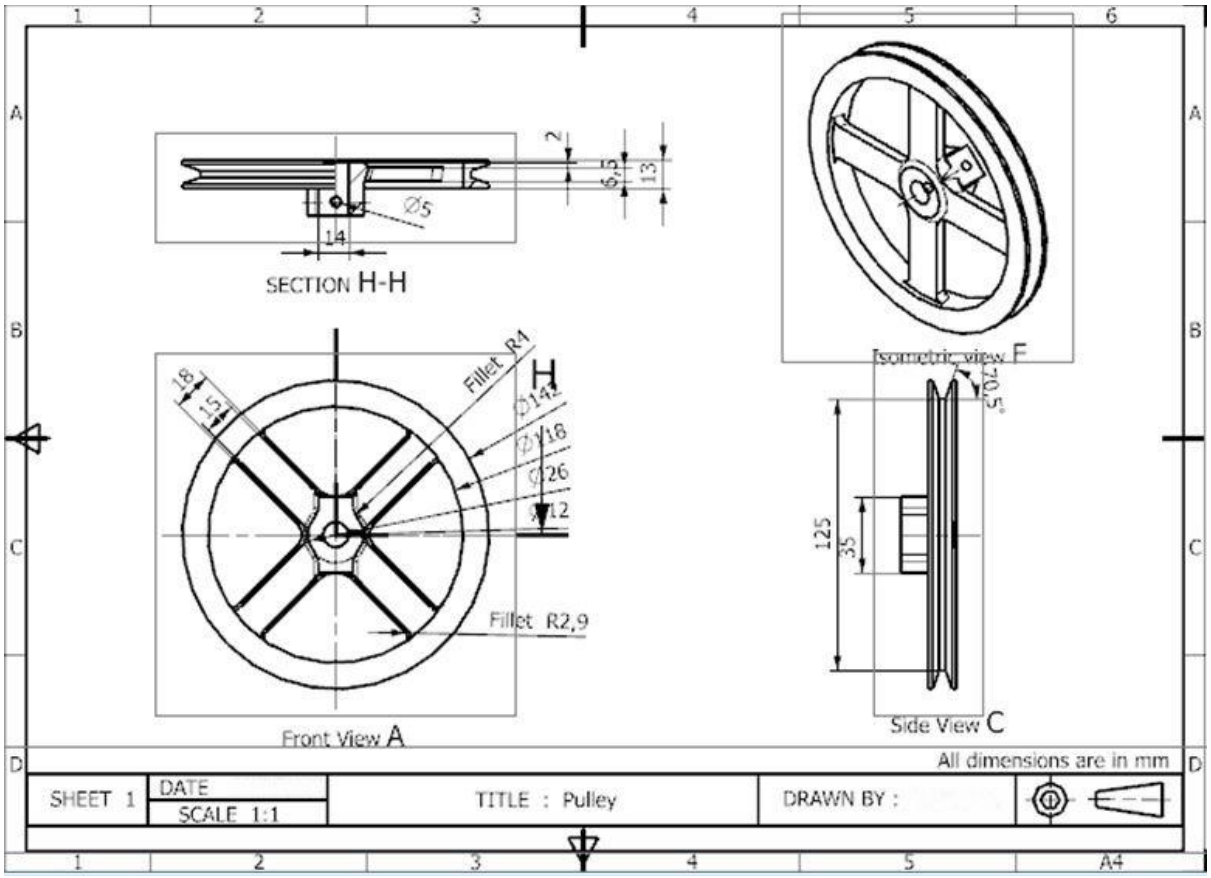
Meshing:

For a 3d component there are two different types of mesh elements, they are

1. Hexa (Brick elements/Rectangular prism)
2. Tetra (Tetrahedron/Triangular pyramid elements)

For the geometry in this analysis, Hexa mesh cannot be used even though the calculation accuracy under hexahedral element is higher than tetrahedral element (Weibing Liu1, 2011) as the geometry cannot be idealized (which affects the mass and

stiffness of the component). Hence Tetra mesh was required. CTETRA (10) is a second order tetra mesh which has 10 nodes when compared to CTETRA(4) which has 4 modes. Meanwhile, calculation accuracy of modal analysis can be improved by increasing the number of nodes because of the order of algebraic equations obtained by discrete differential equation is increased. Hence CTETRA (10) is used which has a higher number of nodes than CTETRA (4). The mesh size has been reduced to 3mm finally after several trials till calculation accuracy has become stabled.



The orthographic and isometric views for the pulley shown above with dimensions

Figure 5.1

5.2 PROCEDURE:

1. Import the 3D CAD geometry file in Siemens Nx11.0 by opening the .prt file.

2. Geometry Idealization is not performed as it usually results in changes to the mass and stiffness of the component which may result in inaccurate results (inaccurate natural frequencies).

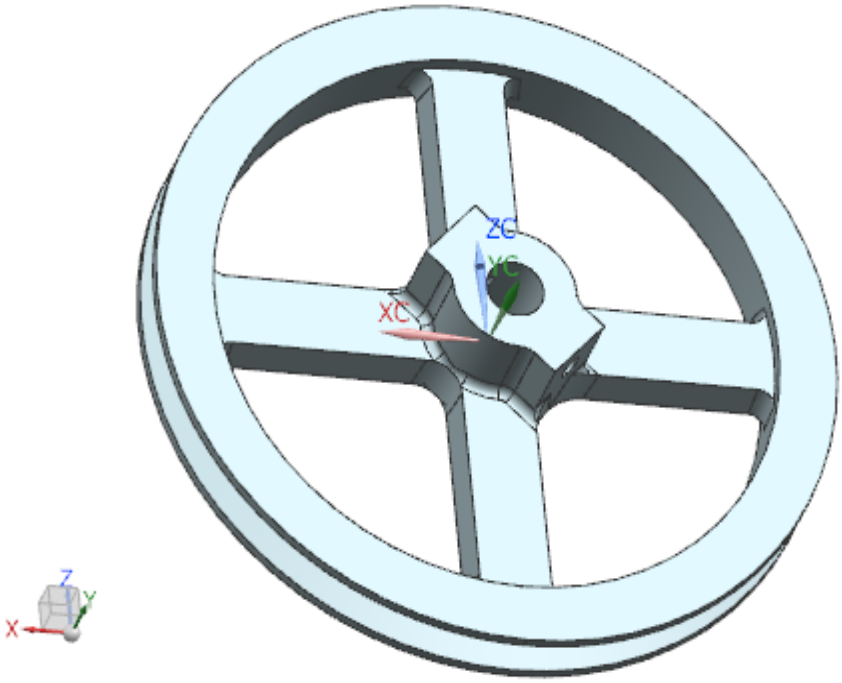


Figure 5.2 CAD model drawing with Siemens NX

5.2.0 Enter Pre/Post Module by creating new FEM and Simulation

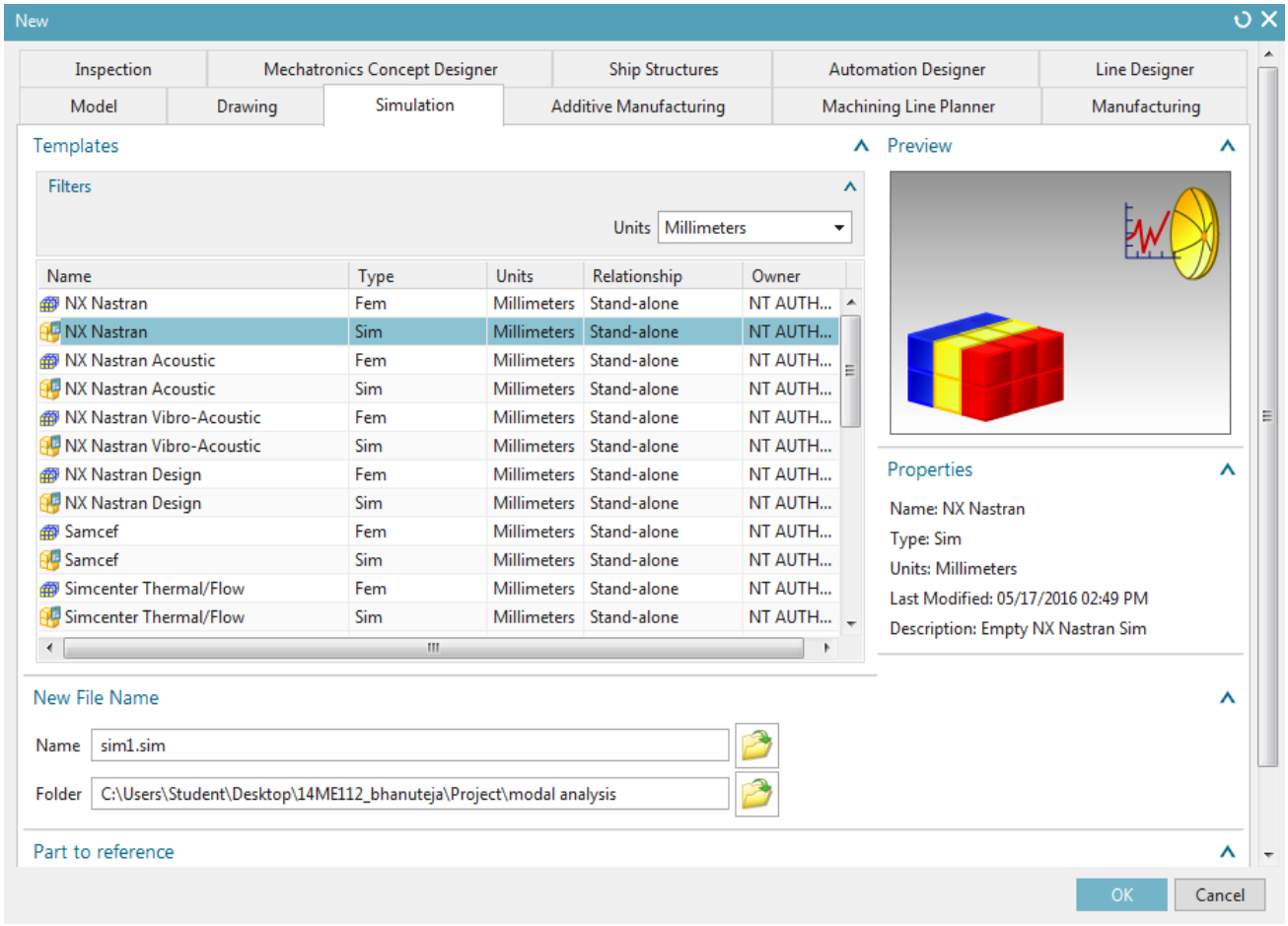


Figure 5.3: Initializing Simulation Interface

5.2.1 An Idealized Part is also created in case the geometry needs modification later.

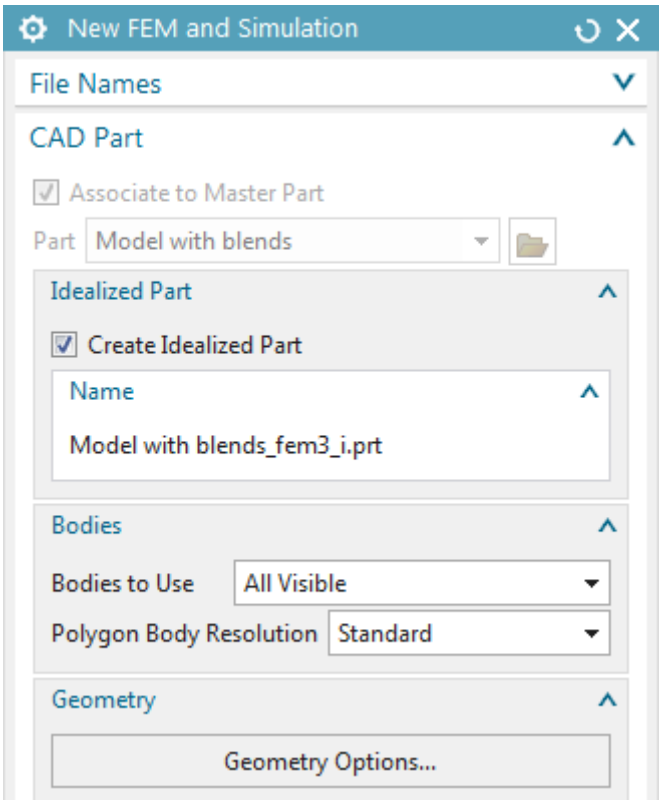


Figure 10 Creating Idealized Part and FEM
Figure 5.4

5.2.2 Create New FEM and simulation

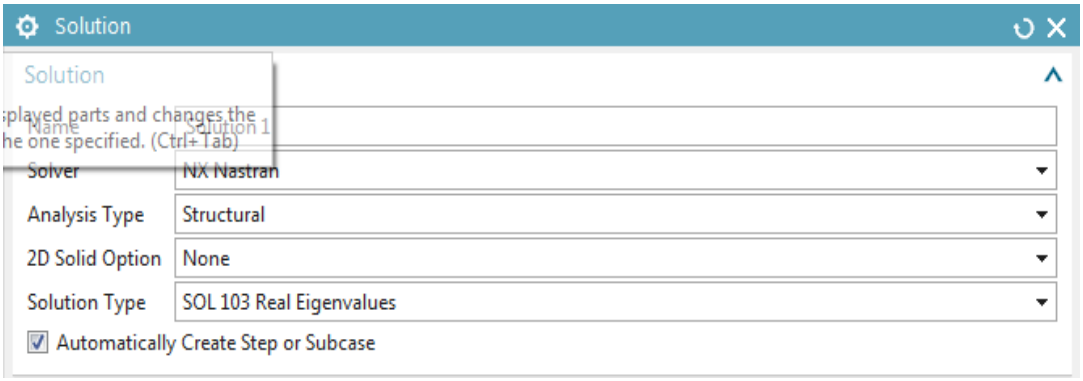


Figure 5.5 Create New solution

3.Simulation module is activated and 103 Real Eigen Values is selected as solution type with the solver as Nastran. and Lanczos data is entered in the solution settings.

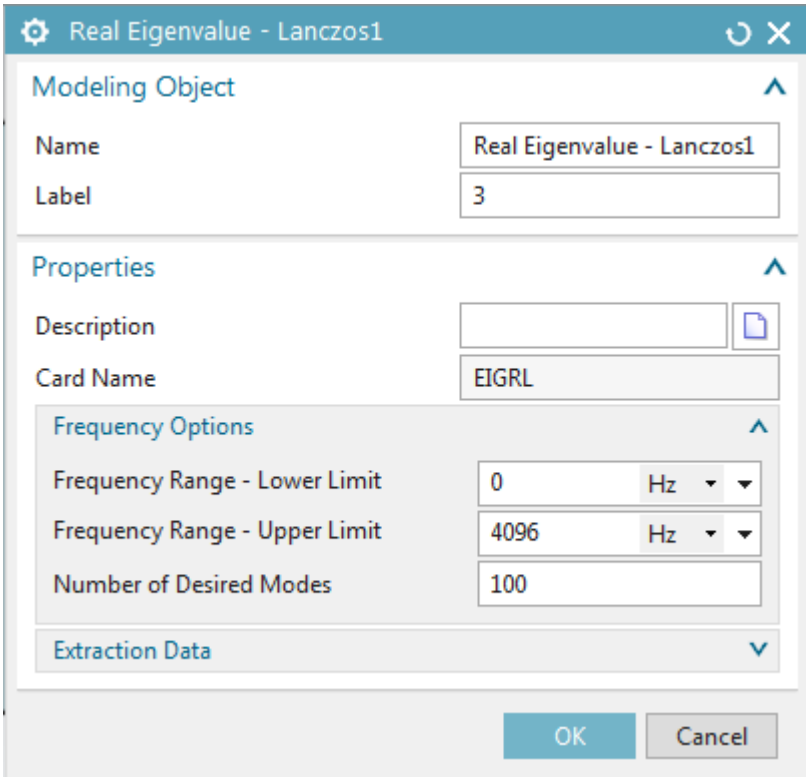


Figure 5.5 Lanczos Data

4. Generate 3D mesh
5. The imported cad geometry contains small blends and hence it is not possible to perform swept hexagonal mesh.

6. The tetra mesh is selected under '3d tetra 'mesh 'CTETRA10' with the element size as 3mm.

7. Mesh size-based Curvature variation as 95 and Element growth rate through volume as 50 with a Small Feature Tolerance 10% of Element size.

8. Material of generated PSOLID1 is Aluminum_A356
9. Material of generated PSOLID1 is Aluminium-A356

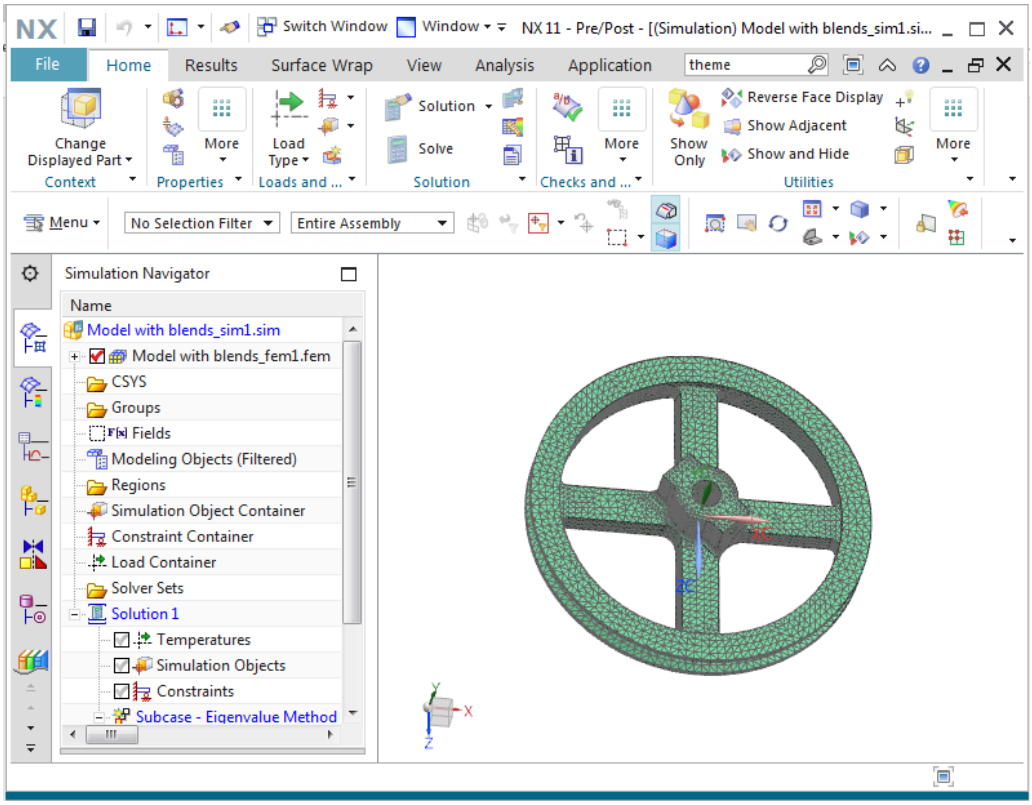


Figure 5.6 Pre-Processing FEM File (meshing) – Simulation Navigator

10. The solution is solved and the normal modes of the component are obtained from post-processing navigator.

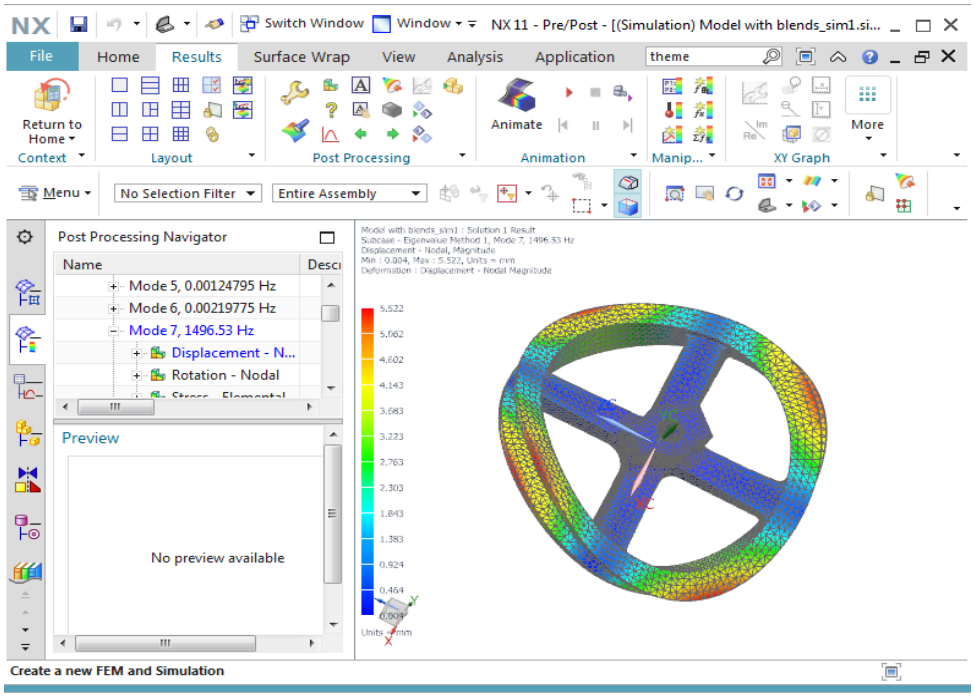
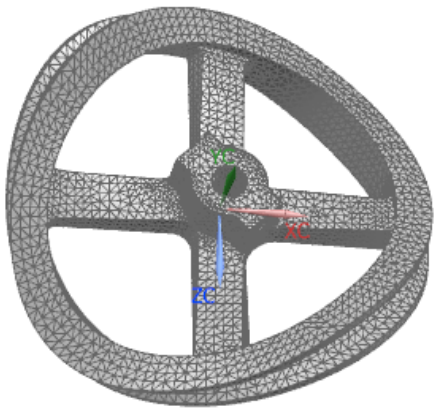
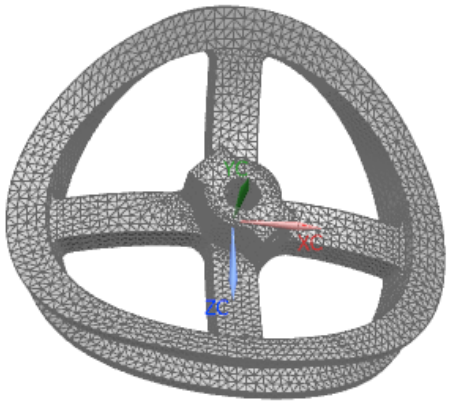
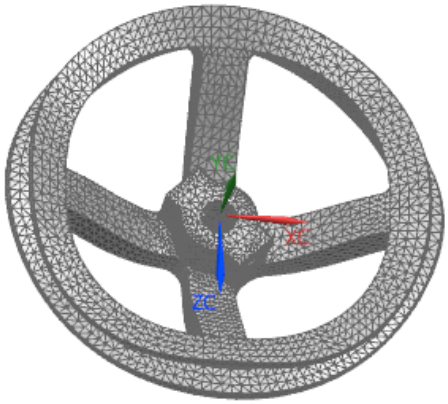


Figure 5.7 Post-Processing Navigator

5.3 RESULTS : Mode Shapes, Natural frequencies

The following are the mode shapes obtained and respective natural frequencies

Modes ranging from 1 to 6 are rigid body motions, hence they are ignored.

Frequency	Mode shape
Mode 7, Frequency 1496.53Hz	
Mode 8, Frequency 1540.4Hz	
Mode 9, 2110.12Hz	

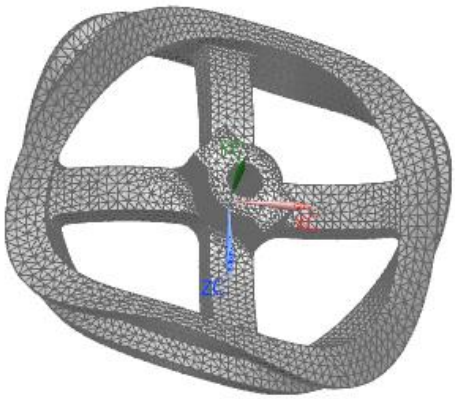
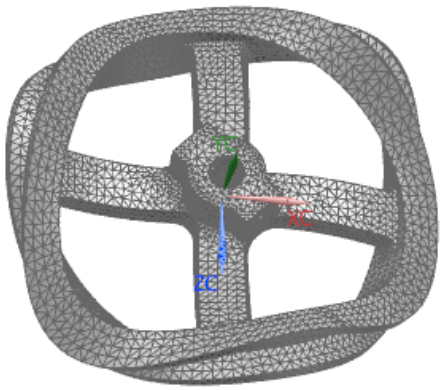
Mode 10, 3587.76Hz	
Mode 11, 3600.12Hz	

Figure 5.8

5.4 SIMULATION DETAILS:

Model Construction

The version of Product used to create this model was NX 11.0.2.7.

Part File names used to construct this model:

File names in session

Part File Name	File Type
Model with blends_sim1.sim	SIM
Model with blends_fem1.fem	FEM

Model Summary:

Solution Summary table

Solution Detail	Status
Solution Name	Solution 1

Solver	NX Nastran
Analysis Linearity	Linear
Solution Type	SOL 103 Real Eigen values

Mesh Summary table

Mesh Item	Count
Total element count	54499
Total node count	85523
Tet10 element count	54499

Meshes

Node Count	*85523*
Element Count	*54499*

Mesh Summary

Name	Type	Element Count	Node Count	Physical Property Name
3d_mesh(1)	CTETRA(10)	54499	85523	PSOLID1

Materials

Materials Used Summary

Name	Type	Referenced By
Aluminum_A356	Isotropic	PSOLID1 (Phys Prop)

Solution & Sub case Details

Solution Details

Solution Name	Solution Description	Solver	Analysis	Solution Type	Eigen value Method
Solution 1		NX Nastran	Structural	SEMODES 103	Lanczos

List by Load

Sub case Name	GLOBAL	Sub case – Eigen value Method 1

Listed by Simulation Object

Subcase Name	GLOBAL

CHAPTER-6

EXPERIMENTAL MODAL ANALYSIS USING SIMCENTER TESTLAB

6.1 INTRODUCTION:

Simcenter Testlab is a complete, integrated solution for test-based engineering, combining high-speed multi-physics data acquisition with a full suite of integrated testing, analytics, and modelling tools for a wide range of test needs.

Impact Testing is ideal for small structures, especially when boundary conditions are seriously disturbed by shaker attachment. While it requires minimum test preparation time, data harvesting is a tedious process. LMS Testlab provides a complete modal testing solution from measuring frequency responses to performing modal analysis using impact hammer frequency response function measurements for simultaneous response channels.

Modal Testing & Analysis:

- Explore root causes of noise, vibration and durability problems with a system dynamic approach modal analysis of the structure.
- It consists of transducers, an analog to digital converter or frontend and host PC.
- It helps to reduce the noise emitted by product.
- It helps in identifying the source of vibrations.

6.2 MINIMAL EQUIPMENT NEEDED:

A minimum of two channels of data must be acquired to measure a structural FRF: an input force and the response output of a test object. In an impact measurement, the input force is provided by a modal impact hammer, while the test object output response is measured with an accelerometer.

Here is a list of required equipment

1. SCADAS.
2. Computer with Simcenter Testlab Impact Testing.

3. Impact hammer.
4. Accelerometer.
5. Appropriate cables.
6. Test object.

6.3 ROVING HAMMER METHOD:

When performing a multiple point modal test with a force impact hammer and accelerometer, there is a choice to move or "rove" either the hammer or the accelerometer between measurement locations.

Roving the impact hammer has some advantages over roving the accelerometer. When moving the accelerometer, one has to unmount and remount the accelerometer at the different measurement points. Moving the accelerometer also changes the mass distribution on the structure, which could alter the natural frequencies.

Moving the hammer is not without considerations however. Because the hammer is a single input and a triaxial accelerometer measures output is three directions, one must take care to ensure the resulting mode shapes are complete.



Figure 6.1: Roving Hammer

6.4 DEFINING GEOMETRY:

In menu bar > Tools > Add-ins > Check in Geometry > OK. This will enable the Geometry in the navigation section.

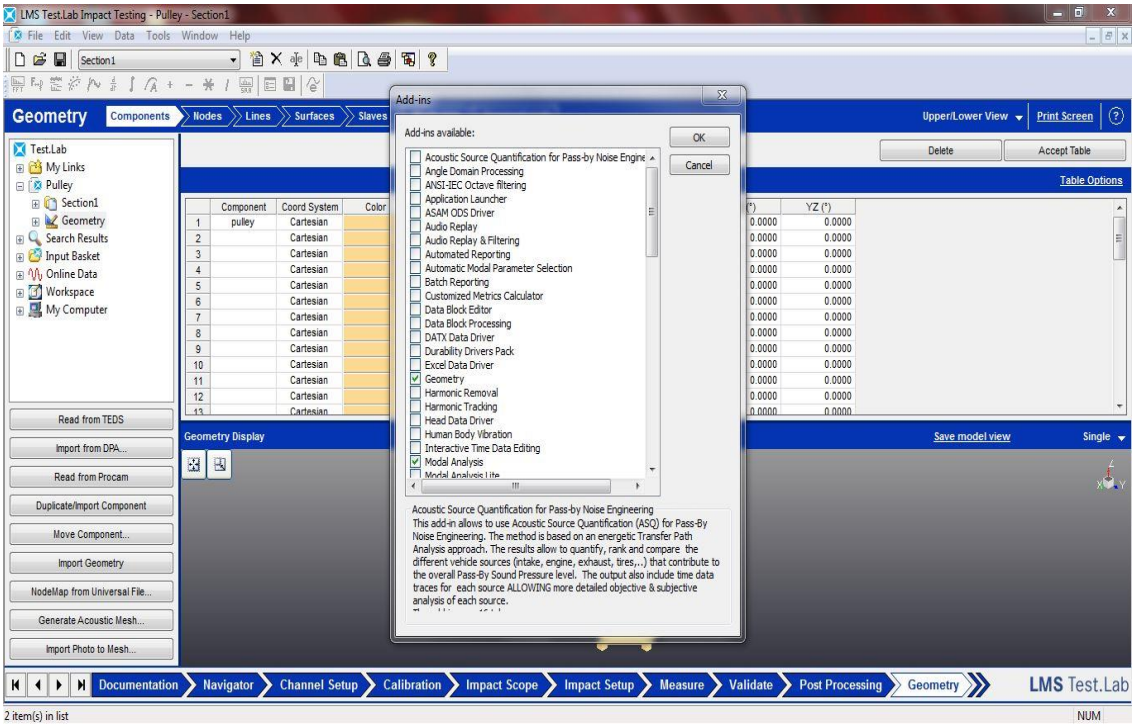


Figure 6.2 : Geometry setup

Steps:

- Name the component as Pulley.
- Enter the coordinates of the reference geometry. During Measure, the software will increment the next point based on the name given. E.g. node 8 to node 9.
- Next connect the points using lines and surfaces.
- Accept the table at the end of each section. Then the reference geometry is created.

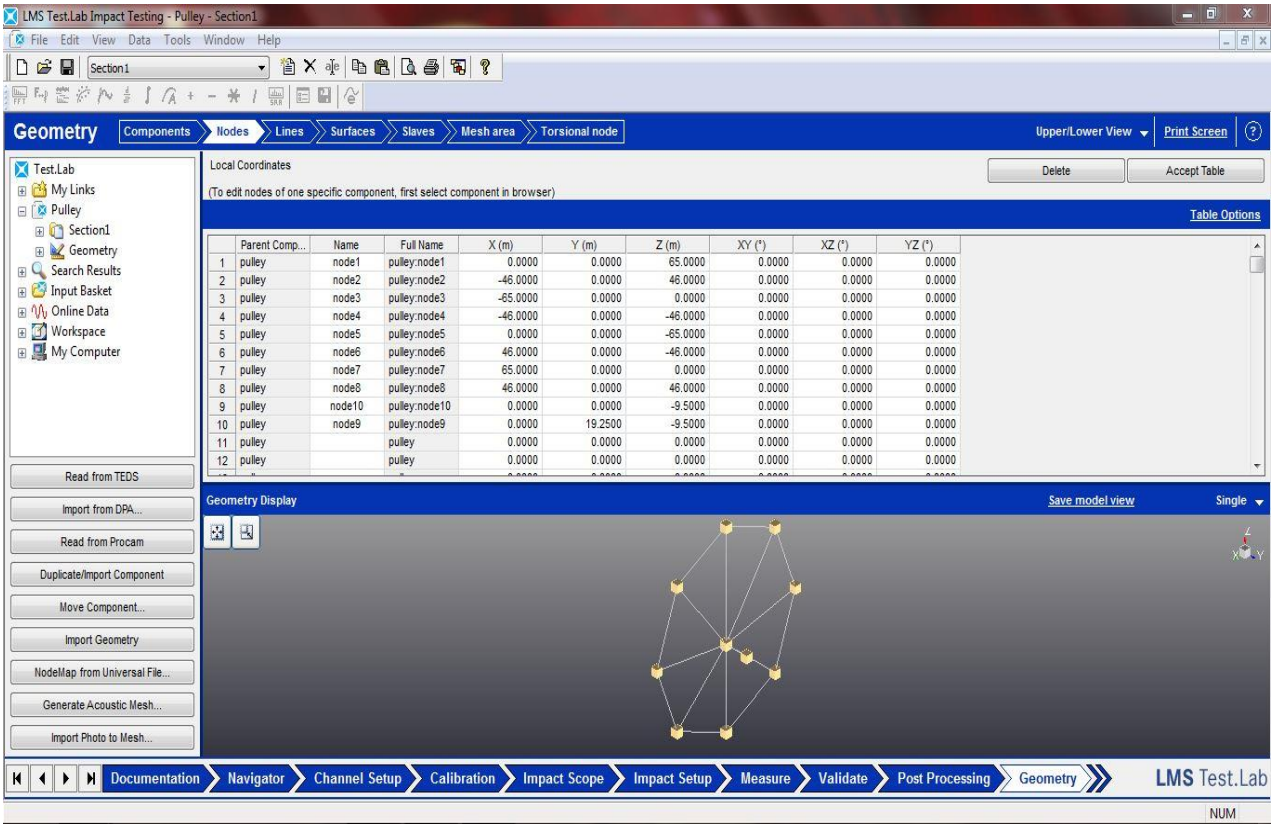
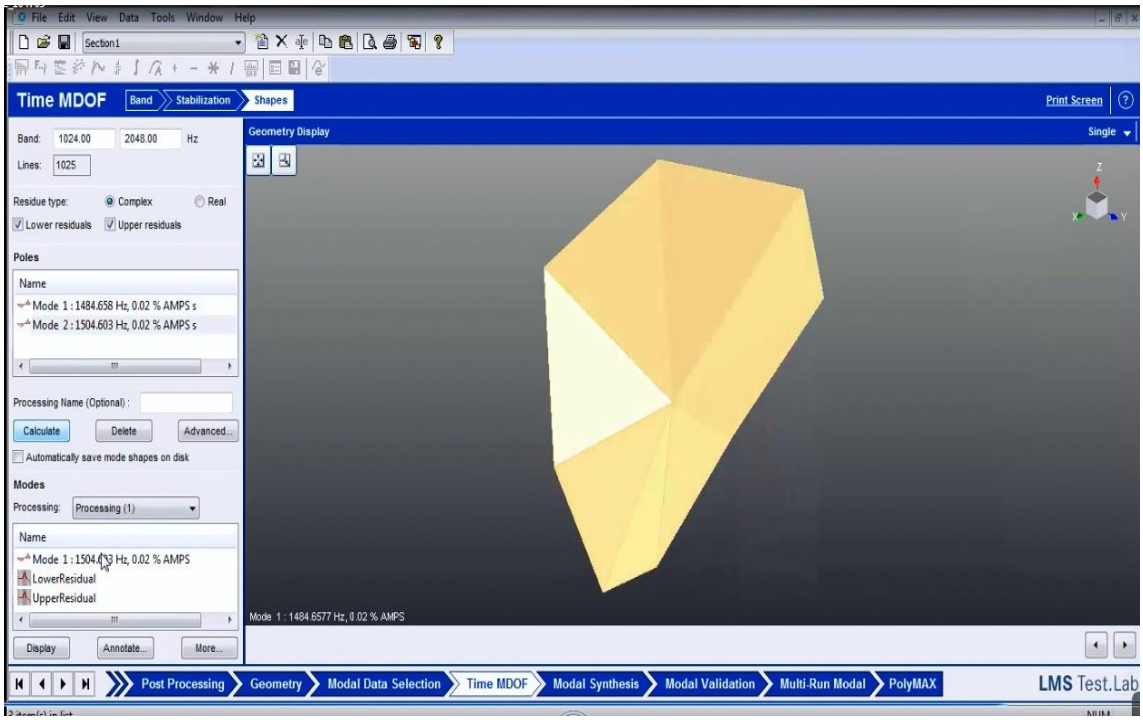


Figure 6.3 : Geometry defining

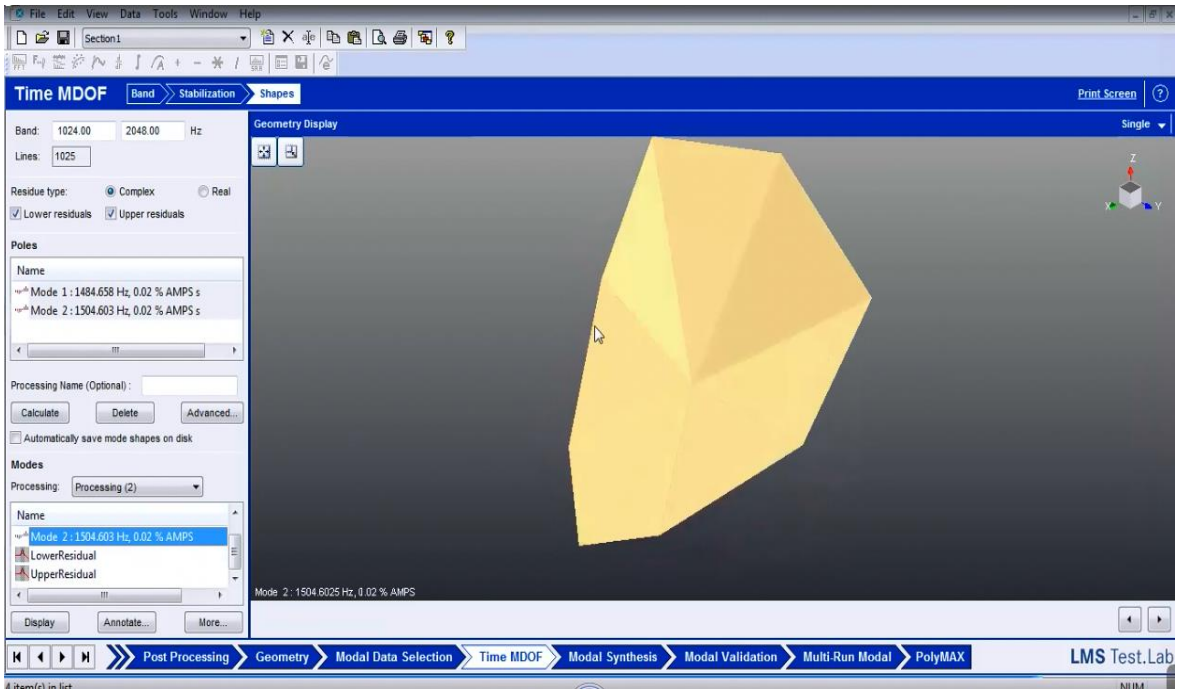
- Now again go to channel setup > Use Geometry > Refresh.
- Select first row of Use Geometry and Input1 and >> Insert. It refers that during measure Input1 (i.e., hammer) will start from given point.
- Follow the above step for accelerometer. Now select the points where the accelerometer is placed >> Insert.

6.5 EXPERIMENTAL RESULTS:

Mode Shape 1 at an average natural frequency of **1484.658Hz**



Mode Shape 2 at an average natural frequency of **1504.603Hz.**

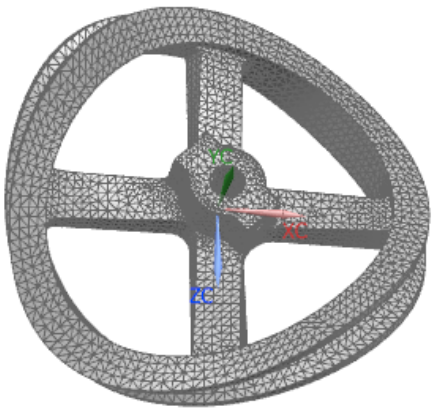
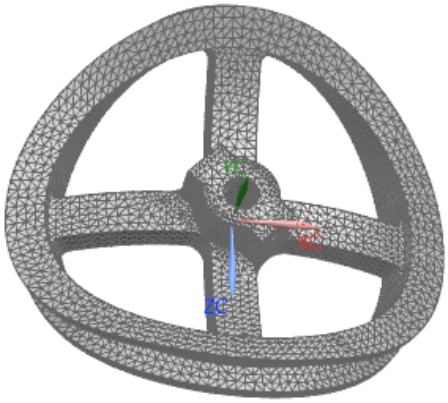


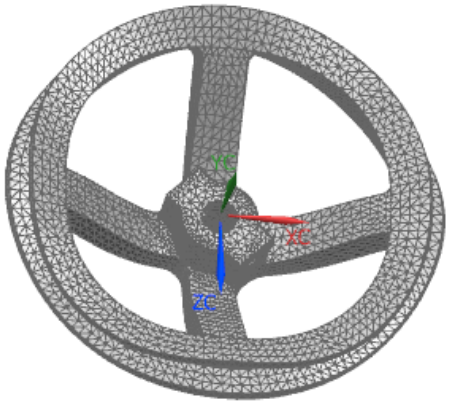
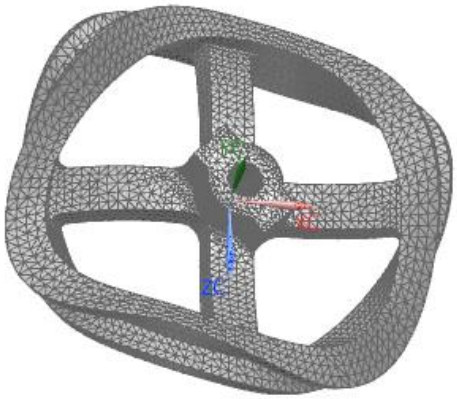
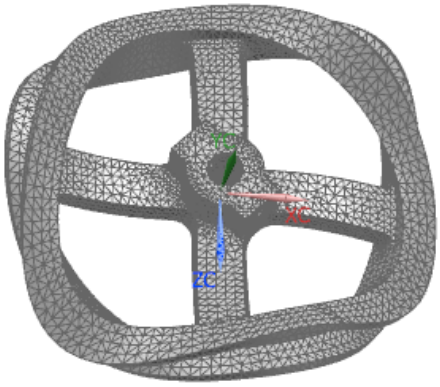
CHAPTER-7

COMPARISON OF NUMERICAL AND EXPERIMENTAL ANALYSIS

7.1 NUMERICAL ANALYSIS RESULTS:

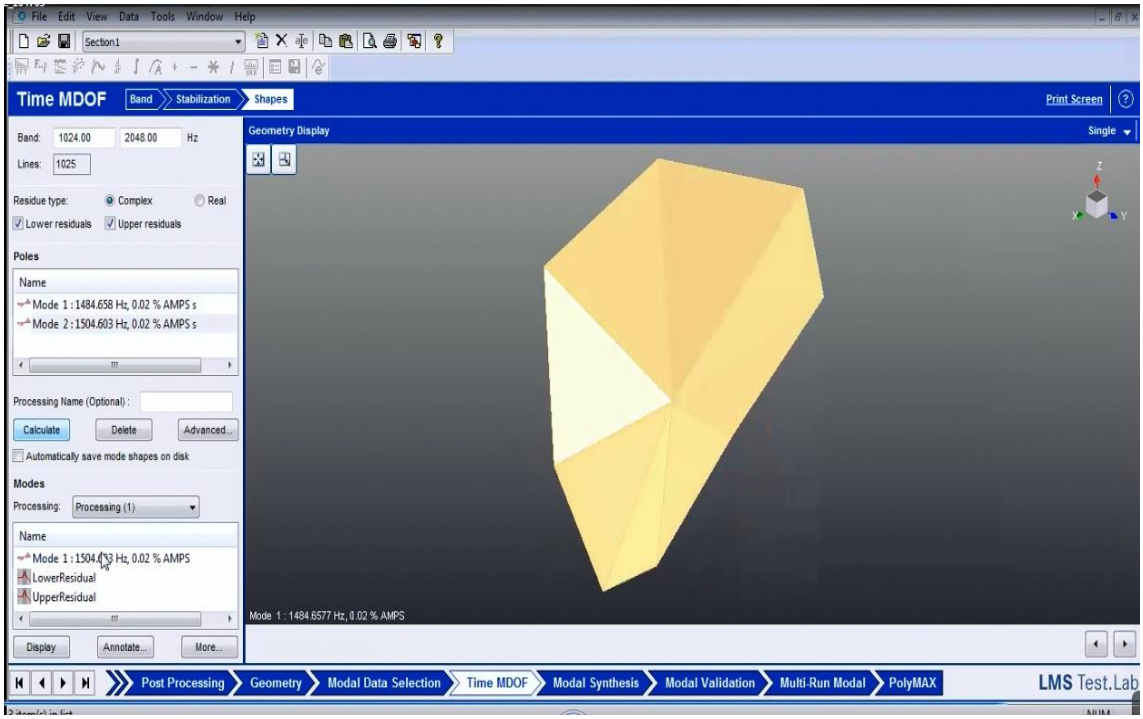
The following are the mode shapes obtained and respective natural frequencies in a bandwidth of 4096 Hz.
Modes ranging from 1 to 6 are rigid body motions, hence they are ignored.

Frequency	Mode shape
Mode 7, Frequency 1496.53Hz	
Mode 8, Frequency 1540.4Hz	

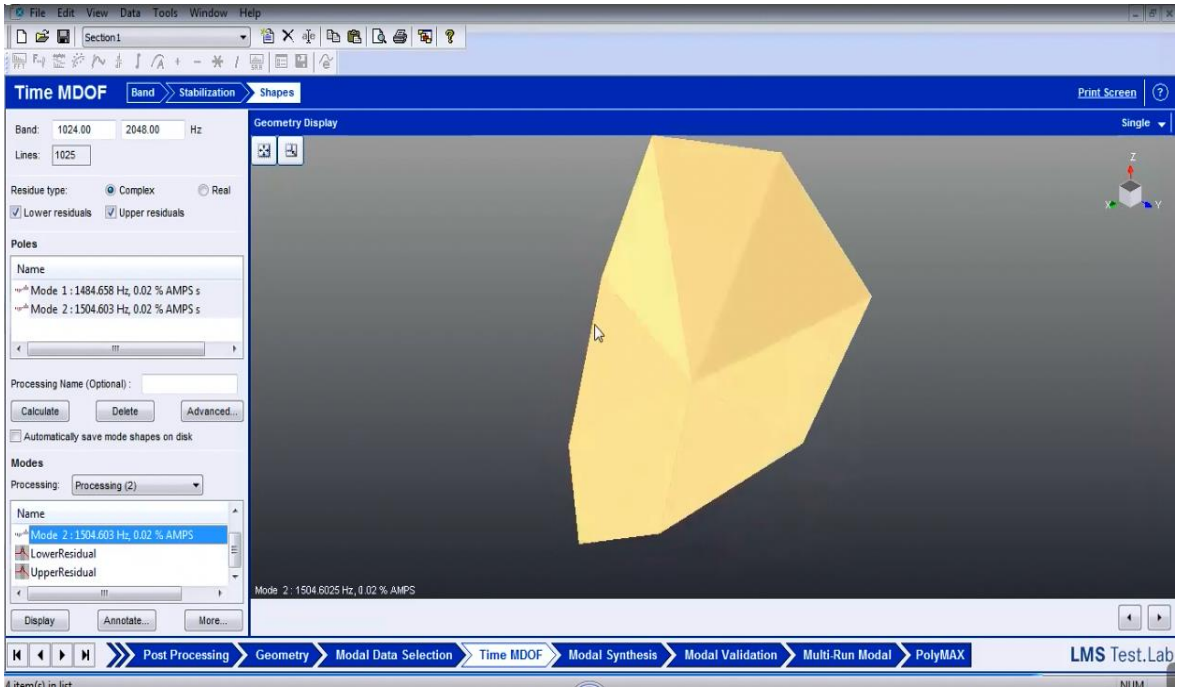
Mode 9, 2110.12Hz	
Mode 10, 3587.76Hz	
Mode 11, 3600.12Hz	

7.2 EXPERIMENTAL RESULTS:

Mode Shape 1 at an average natural frequency of 1484.658Hz



Mode Shape 2 at an average natural frequency of 1504.603Hz.



7.3 COMPARISON:

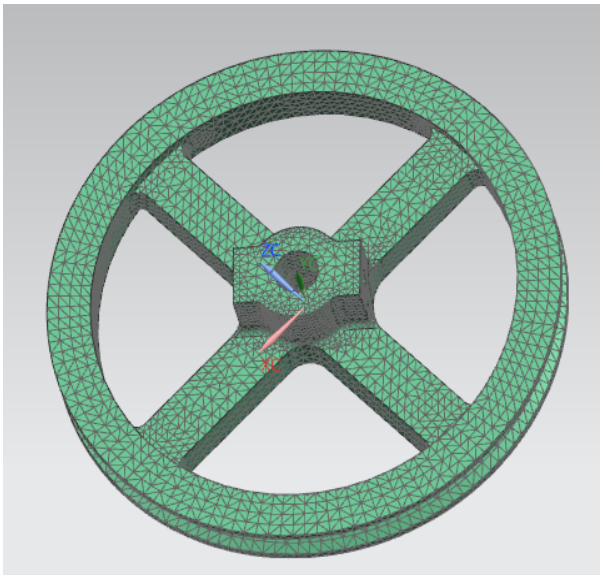


Figure a

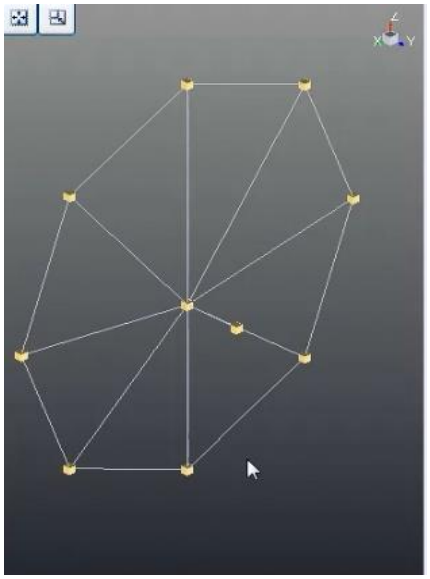


Figure b

For the first normal mode obtained by solving with siemens NX_11 simulation software using Nastran as solver was 1496.53Hz. First normal mode obtained from the impact testing results was 1484.658Hz. Similarly, there was a variation in natural frequency in second mode.

Difference in natural frequencies in both of the analyses is because of:

- The difference in geometry in CAD model and physical component.
- External factors affecting the experimental analysis such as air flow, noise, unwanted excitation on test rig, inaccurate triggering on points etc.
- The operational stresses induced in the component.
- Loss of material (mass) due to friction between belt and pulley.
- Dirt and debris accumulated over the component.
- As the pulley is an unfinished casted component, there is a mass difference between the designed and output component. The internal structure is not considered.

CHAPTER-8
CONCLUSION

An accurate Cad model of the component was drawn in Siemens part modelling. Pre-processing and meshing were performed in Siemens NX, the type of mesh used was CTETRA (10) by using seed mesh technique. To determine the mode shapes, the modal analysis was performed using the solution type 103 Eigen values using Nastran as solver. Mode shapes were obtained by solving it by defining the desired output range from 0 Hz-4096 Hz in Lanczos data. Hence the natural frequencies and mode shapes of the component were obtained from the Post-processing navigator in the Siemens NX, and the results were validated from the results obtained from Experimental modal analysis using Simcenter testlab. The results obtained in both cases were very close with a variation of ±12 Hz.

To conclude our work, we compare the functional frequency (or working frequency) of the washer pulley with that of the experimental results. The working frequency of washing machines will be 1200-1800 RPM i.e., is 23.33Hz – 30Hz. From this we can conclude that, the washer pulley is working in safe frequency and will not reach resonance.

References:

1. Weibing Liu1, a. M. (2011). Effect of Mesh Size of Finite Element Analysis in Modal Analysis for Periodic Symmetric Struts Support . *Key Engineering Materials,Trans Tech Publications, Switzerland*.

2. Schrötter, Martin & Hagara, Martin & Lengvarský, Pavol. (2014). EXPERIMENTAL MODAL ANALYSIS OF WASHING MACHINE PULLEY.